

KENYATTA UNIVERSITY
DEPARTMENT OF BOTANY
MODULE SBT 201/207: PLANT
FUNCTION / INTRODUCTION TO
PLANT PHYSIOLOGY

LESSON 1: SOLUTIONS

1.1 Introduction

In Cellular Basis of Life you learnt that the basic structural and functional unit of life is the cell. Substances from outside and those produced within the cell are dissolved in water to form solutions. The water and solutions move into and out of the cell. Solutions are found in the cytoplasm, cell vacuoles and within cytoplasmic organelles inside the cell. Outside the plant system solutions are found in the soil as soil solutions and water vapour is found in the atmosphere. Cellular reactions take place in solutions within the cell. Solutions are therefore important to the functioning of plants. Their properties are affected by the pH.

1.2 Objectives

By the end of this lesson, the students should be able to:

- a) explain the composition of a solution.
- b) identify the properties of a solution
- c) determining the hydrogen ion concentration and pH of solutions
- d) prepare some solutions.

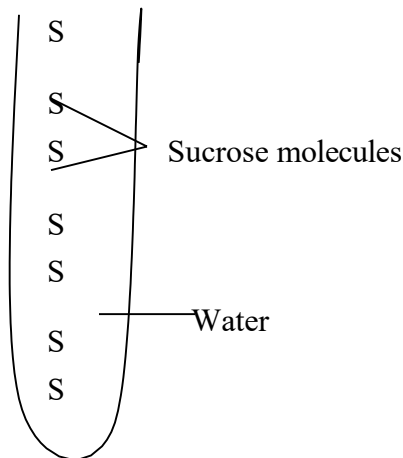
1.3 Content

1.3.1 Definition

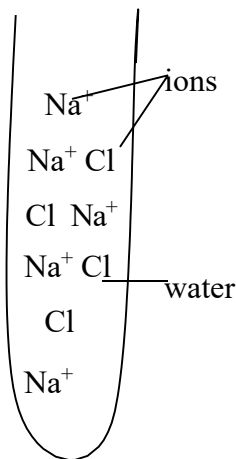
A solution is a mixture of two or more substances in which particles of one substance are separated from one another and dispersed throughout the medium of the other. The

particles are either molecules or ions. A solution is also known as a crystalloid.

Common examples of solutions are those of sucrose dissolved in water and salt (sodium chloride) dissolved in water.



Sucrose is a nonionic substance and so remains intact in water.



Salt (sodium chloride) is an ionic substance. When placed in water it undergoes ionization and dissociates to sodium and chloride ions.

The dispersed substance in a solution is called a solute whereas the substance in which the solution particles are dispersed or dispersion medium is called the solvent. The solute remains evenly dispersed in a solution and does not settle out no matter how long the solution lasts. The amount of solute in a solvent gives the concentration of the solution and gives rise different types of solutions.

1.3.2 Types of solutions.

A dilute solution is obtained when a small amount of solute is added to a solvent. A saturated solution is one that contains the highest amount of the solute dispersed in a solvent, at a given temperature and pressure. If more solute is added to the saturated solution it settles down. This excess solute dissolves when the solution is heated and will continue being in solution even at a lower temperature, giving rise to a supersaturated solution. If agitated or more solute is added to it the excess amount of solute settles down leaving a saturated solution.

The concentration of a defined amount of solute in a defined amount of solvent may be expressed as molarity, molality, normality, percentage and equivalent.

a) Molarity:

A molar (1M) solution is obtained when weight of a substance equal to its molecular weight in grams is dissolved in water and the solution is made up to one litre e.g. 1M of sucrose contains 180.6g in 1 litre of solution. 1M solution of a substance contains Avogadro's number of solute molecules in one litre of solution.

b) Molality

A molal (1m) solution is formed when the molecular weight of a substance in grams is added to one litre of water and dissolved. The final volume of the solution will be a little more than one litre. Equal volumes of molal solutions contain the same number of solvent molecules but may not contain the same number of solute molecules.

c) Normality

A normal (1N) solution or equivalent solution is formed when the equivalent weight of a substance in grams is dissolved in a solvent and the solution made to one litre. 1.008 grams of hydrogen. The gram equivalent weight of a substance can be obtained by dividing its molecular weight by its valency.

d) Percentage solution.

A percent solution is obtained when 1 gram of substance is dissolved in 99 grams of water. Percentage solutions are prepared by dissolving definite weights of substances in weights of the specific substance.

To prepare all these solutions grams are used. Lower weights can be used e.g. milligrams that give millimolar, millimolal and milliequivalent solutions. Micrograms gives micromolar, micromolal or microequivalent solutions. Percentage solutions remain the same even if lower weights are used.

1.3.3 States of matter.

With all the three states of matter, theoretically nine types of solutions are possible.

	Solute		
Solvent	Liquid	Solid	Gas
Liquid	Liquid in liquid	Solid in liquid	Gas in liquid
Solid	Liquid in solid	Solid in solid	Gas in solid
Gas	Liquid in gas	Solid in gas	Gas in gas

Of all these types of solutions as per states of matter, the commonest ones are those of liquid as solvent, particularly water and solid, liquid and gas as solutes.

Factors of temperature and pressure affect the solubility solid, liquid and gas in a liquid. An increase in temperature increases the solubility of a solid and liquid in a liquid but decreases the solubility of a gas in a liquid. An increase in pressure increases the solubility of gases in liquids according to Henry's law. The law states that if there is no chemical reaction between a gas and a liquid, the amount of gas absorbed in the liquid will be proportional to its partial pressure.

A solution of liquid in a liquid may be perfectly miscible, partially miscible or immiscible. In a perfectly miscible situation, one liquid is completely dissolved in the other liquid of greater amount or solvent. Partially miscible liquids will mix up when

shaken but their immiscible portions will separate out on standing again. Immiscible liquids will not mix at all when shaken but forms emulsions.

1.3.4 pH of solutions.

pH is the negative logarithm of the hydrogen ion concentration. Hydrogen ion concentration is used to determine the basicity and acidity of a solution. An acid is a molecule or ion that can donate a proton (H^+) to any other molecule or ion. A base is any molecule or ion that will accept a proton.

$$pH = -\log(H^+)$$

pH stands for “potential of hydrogen” There is a pH scale that covers a range of values from 0 to 14. These values have been derived from the hydrogen ion concentration expressed in terms of normality. For example a litre of pure water has 0.0000001N or 10^{-7} . Its negative logarithm of hydrogen ion concentration is calculated as follows:

$$\begin{aligned} pH &= -\log 10^{-7} \\ &= \log \frac{1}{10^{-7}} \\ &= 7 \end{aligned}$$

The pH of pure water is therefore 7 and is considered neutral. pH values below 7 give acid solutions whereas those above 7 give basic solutions. pH values differ by a factor of ten.

1.3.5 pH scale

Hydrogen ion Concentration (In terms of normality.)	Exponential form	pH value
1	10^0	0
0.1	10^{-1}	1

0.01	10^{-2}	2
0.001	10^{-3}	3
0.0001	10^{-4}	4
0.00001	10^{-5}	5
0.000001	10^{-6}	6
0.0000001	10^{-7}	7
0.00000001	10^{-8}	8
0.000000001	10^{-9}	9
0.0000000001	10^{-10}	10
0.00000000001	10^{-11}	11
0.000000000001	10^{-12}	12
0.0000000000001	10^{-13}	13
0.00000000000001	10^{-14}	14

Hydrogen ion concentration of solutions changes when an acid or a base is added to them. There are solutions that resist changes in pH when small amounts of acid or base are added. These are called buffer solutions. A common buffer is that of a mixture of acetic acid and sodium acetate. In biological systems many amino acids and proteins operate as primary buffers.

1.4 Summary.

A solution is a mixture of two or more substances in which all the molecules or ions of one substance are separated from one another and dispersed throughout the medium of the other. Ionic and monionic substances can be used to prepare solutions. Their concentrations in a fixed amount of solvent under specified conditions gives different types of solutions; dilute, saturated and supersaturated. Relative weights of solutes and defined amounts of solvent gives solutions whose concentrations are expressed by molarity, molality, normality and percent solutions.

In a broad sense, solutions can be formed according to the three states of matter. Theoretically, nine possible combinations can be obtained. The commonest ones are

those of solids and liquids in liquids. The basicity and acidity of these solutions can be expressed using hydrogen ion concentration. This is given as pH that ranges from 0 to 14. These values are the negative logarithm of the hydrogen ion concentration.

1.5. Activity

Preparation of different types of solutions:

- a) Prepare 50 ml of a 5 percent by weight solution of sucrose in water.
- b) Prepare a 10 percent ethanol solution
- c) Prepare 100 ml of 1 normal sodium chloride solution.
- d) Prepare 100 ml of 0.5 molar sodium chloride solution.

1.6 Questions:

1. Define the following solutions;

- a) Dilute solution (b) saturated solution (c) supersaturated solution (d) normal solution.
2. Write short notes on buffer solutions.
3. Distinguish between the following (a) acid and base. (b) molarity and molality.
4. Contrast the behaviour of sucrose and sodium chloride when they are dissolved in water.
5. What is a solution?

6. Discuss the theoretical possibilities of solutions formed from all the three states of matter.

7. State Henry's law.

8. What is hydrogen ion concentration?

LESSON 2: COLLOIDS

2.1 Introduction

In Lesson 1 we learnt that solutions are formed when solute substances are dissolved in solvents. In this lesson we look at another system formed in more or less the same manner. Materials are also dissolved in the cell protoplasm (cytoplasm and nucleus) much of the particulate phase of the protoplasm is colloidal in nature. The protoplasm is referred to as a colloidal complex. This state is due to the presence of the proteins. The proteins that are dispersed in the protoplasm provide an immense surface area. This produces the necessary conditions for adsorption and chemical movement that makes it possible for the reactions of life to take place.

2.2 Objectives

By the end of this topic, the students should be able to

- a) explain the composition of colloidal systems.
- b) identify the properties of colloidal suspensions.
- c) explain the polyphasic nature of the protoplasm that is a colloidal system.
- d) contrast emulsions and suspensions with both colloids and solutions

2.3 Content

2.3.1 Composition

The term colloid is derived from two Greek words “kolla” which means “glue” and “eidos” which means “like”. The term colloid was used to describe glue-like preparations for example solutions of certain protein.

The size of colloidal particles are intermediate between those of solutions and suspensions. The particles of colloids range from 1 to 200nm. They are larger than the size of most molecules. All states of matter can be involved in the formation of colloidal suspensions. The commonest one is the dispersion of a solid in a liquid. The dispersion medium may be a liquid, gas or solid. Some examples are: smoke is a solid dispersed in gas. Milk and mayonnaise are a liquids dispersed in a liquid. Pumice stone is a gas dispersed in a solid.

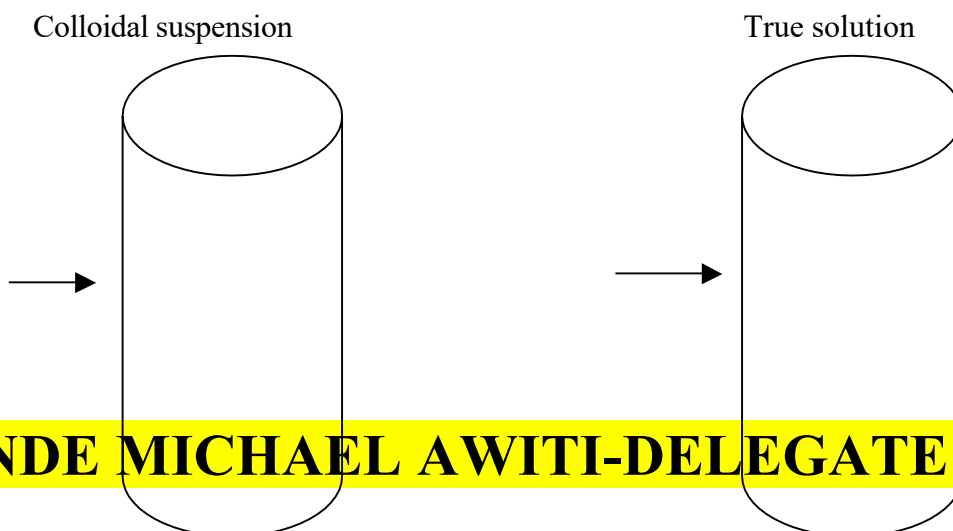
Generally a colloidal system exists in either of the two forms, sol and gel. A sol has a high degree of fluidity and appears like a solution. A gel on the other hand appears as a solid or semi solid and is more or less an elastic system. A colloidal system can change from sol to a gel, a process known as gelation. The reverse change from gel to sol is known as solation.

Colloidal dispersions in liquids are of two general classes, lyophilic and lyophobic systems. In the lyophilic systems the dispersed phase and the liquid dispersion medium are attracted to each other. In lyophobic systems there is no affinity between the dispersed and dispersion medium. If the dispersion medium is water, these two systems are known as hydrophilic and hydrophobic respectively. There is an affinity of the dispersed phase towards water in a hydrophilic system e.g. starch, agar, gum and cellulose. In hydrophobic system there is no affinity towards water by the dispersed phase e.g metals and their sulphides and hydroxides.

2.3.2 Properties of colloids

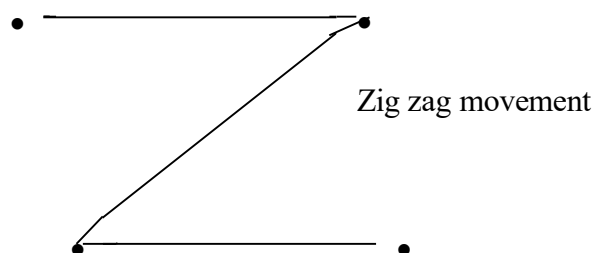
1. Tyndall effect.

If a narrow beam of light is passed through a colloidal suspension and viewed at right angles, a visible path will be observed. This phenomenon is called Tyndall effect. This is due to the scattering of light by large particles of the dispersed phase. If the light beam is passed through a true solution the Tyndall effect is not observed because the particles of the dispersed phase are too small to scatter the light.



2. Brownian movement.

This type of movement was observed by Robert Brown (1828) in pollen grains that were kept in a drop of water and observed under a microscope. There was a rapid oscillatory movement which was later observed in colloidal particles. Colloidal particles in a suspension move in an irregular random fashion or zig zag manner.



The motion is caused by the uneven bombardment of the colloidal particles by molecules of the dispersion medium. Being small enough, the colloidal particles are moved by molecules in the direction of least resistance. The direction constantly changes. This type of movement cannot be observed in a true solution.

3. Filtration (Filterability).

Colloidal particles range in size from 1 to 200 nm. Therefore, they cannot be separated out from the dispersion medium with an ordinary filter paper. Their separation can be achieved by using ultrafilters or millipore filters. These filters have pore sizes ranging from 10nm to 5 μ and have been constructed from biologically inert cellulose esters. Components of a true solution cannot be separated even by ultrafilters. Colloids can be separated even by ultrafilters. Colloids can be separated from a mixture of solutions and colloids by a process known as dialysis using a parchment paper. Water and a true solution diffuse through the parchment paper but not the colloids.

4. Adsorption:

This is the tendency of molecules or ions to adhere to the surface of certain solids or liquids. The capacity of adsorption depends on the amount of surface exposed and the

chemical nature of the constituents involved. Colloidal systems in the living cells provide immense adsorptive capacity for reactants.

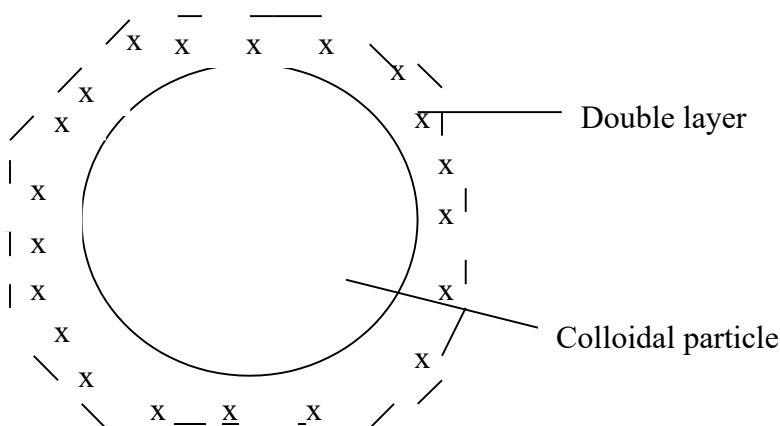
5. Viscosity

Is the resistance of a liquid to flow. Colloids show viscosity and they are more viscous than solutions.

6. Electrical properties

Colloidal particles carry electric charges. The charge is the same for all particles of a given colloidal system. This property is mainly responsible for the stability of colloidal suspensions. The electric charges on colloidal particles are either positive or negative. Like charges repel each other and so they cannot collide, aggregate and precipitate.

Two examples can be used to illustrate this property; ferric hydroxide and arsenic sulfide. Ferric hydroxide gives Fe^{3+} and Cl^- . Particles carry positive charges forming a layer of these charges. The free Cl^- are attracted and accumulate secondarily around the positive charges forming a double layer.



Positive charges of the dispersed phase gather at the cathode when an electric current is passed through a colloidal suspension. The negative charges collect at the anode in an electric field. This is a process originally called cataphoresis but now known as electrophoresis.

7. Flocculation

This process is also known as precipitation. When the double layer formed in a colloidal suspension that is responsible for stability of a colloidal suspension is destroyed or reduced, the particles collide, aggregate and finally precipitate. The destruction or reduction of a double layer removes or reduces the difference of electrical potential across the double layer. The particles settle down in the system giving rise to a phenomenon called flocculation.

Flocculation can be achieved by adding an electrolyte e.g. HCl to a colloidal suspension e.g. arsenic sulfide. The extent to which an ion will cause flocculation depends on its valence. The lower the valence the less efficient the ion. Flocculation in a colloid can be prevented by the presence of another colloid e.g. gelatin protects silver halides in photographic plates.

2.3.3 Protoplasm

In the cell protoplasm, watery solutions of salts form the continuous phase whereas emulsions of organic acids, fats and proteins form the discontinuous phase. It has the properties of both suspensions and emulsions.

Many things are dissolved in the protoplasm, thus it is described as a polyphasic colloidal system. The dispersed colloidal particles are often hydrated by a shell of water molecules. It behaves as a hydrophilic sol. The protoplasm, however, changes repeatedly from sol to gel and gel to sol as the ratio of water and organic compounds fluctuates. It is mostly in a permanent gel stage.

The fact that it is a lyophilic colloid, the protoplasm exhibits stability. This is caused by the attraction between the colloidal particles and water molecules.

2.3.4 Emulsions.

Emulsions are prepared from two immiscible liquids when they are shaken together. Small droplets of one liquid (dispersed phase) are dispersed throughout the other liquid (dispersion medium). The particles of the dispersed phase are more than $0.1\text{ }\mu\text{m}$ in diameter. These particles or small droplets have a tendency to coalesce and form larger and larger droplets until the two liquids separate into two distinct layers. Examples are Homogenized Milk, Mayonnaise, Crema (type of cream), Vinaigrette (type of salad dressing), Butter, Margarine, Cold Cream (cream with cooling feeling on skin) and Emulsion Paints.

2.3.5 Suspensions

A suspension is also referred to as a coarse dispersion. It is formed when particles of matter are dispersed throughout the medium but do not separate into molecules. The size of the particles is more than $100\text{ }\mu\text{m}$ in diameter and can be filtered out by a filter paper. Suspensions are unstable system because after some time the dispersed particles settle down and the liquid becomes free of them. Examples are Muddy water, Milk of magnesia, Sand particles suspended in water, Flour in water, Slaked lime for whitewashing, Paints in which dyes are suspended in turpentine oil.

2.4 Summary

Colloidal systems consists of a dispersed phase and dispersion medium. They are formed from all states of matter. Attraction or repulsion can exist between particles and dispersion medium. Colloidal particles scatter a beam of light, move in a zig zag manner and can be filtered out. They are responsible for higher viscosity, high adsorptive capacity and observed electrical properties. In a living system, the protoplasm exists as a polyphasic colloidal system.

2.5 Activity.**a) Preparation of a colloid.**

Place ordinary clay soil in water. Agitate thoroughly to obtain a murky liquid of uniform colour. Allow the mixture to stand for along time. Large and small particles will settle out. The liquid part containing very minute suspended particles, clay micelles is the colloid.

b) Brownian movement

Mix a drop of Indian ink with 5 drops of water. Mount one drop of the mixture on a microscope slide. Place a coverslip on top of the drop and examine under the microscope using low power then high power. Remove the slide, warm slightly over a candle or lit matchstick. Clean the lower part of the slide and re-examine under the microscope.

2.6 Questions.

- a. Explain how flocculation in a colloid can be prevented.
- b. Define the following terms. (a) adsorption (b) viscosity (c) double layer (d) dialysis (e) electrophoresis.
- c. Briefly describe an experiment used to illustrate brownian movement.
- d. What is the tyndall phenomenon?
- e. Explain how stability of a colloidal suspension is achieved.
- f. Distinguish between the following.
 - i) lyophilic and lyophobic systems
 - ii) solation and gelation.
 - iii) dispersed phase and dispersion medium.
 - iv) hydrophobic and hydrophilic systems.
- g. Differentiate between a suspension and an emulsion.

LESSON 3: DIFFUSION, IMBIBITION AND OSMOSIS.

3.1 Introduction

In this lesson you will learn the movement of molecules in response to concentration gradients and how hydrophilic particles attract and hold water molecules.

3.2. Objectives.

By the end of this lesson you should be able to:

- a) distinguish between diffusion and osmosis.
- b) appreciate the role of hydrophilic particles in the hydration of plant cells.

3.3. Content

3.3.1. Diffusion.

Molecules of gas or solute in solution are continuously in motion. The net movement from one point to another because of the random kinetic activities of the molecules or ions is called diffusion.

Diffusion occurs in response to difference in concentration (concentration gradient) of substances between one point and another. Examples of a concentration gradient in living cells: as sugar in the cytosol is taken in and metabolised by a mitochondrion, its concentration near the mitochondrion is lower than its concentration near a photosynthesizing chloroplast in the same cell.

The rate of diffusion is proportional to:

- a) The kinetic energy of the molecules ie. their temperature).
- b) Their size: diffusion rate is proportional to the square root of the molecular weight.
- c) The density of the medium through which they pass.
- d) The gradient of the concentration over which they diffuse.

When uniform distribution of molecules occurs, a dynamic equilibrium is established and net movement of molecules ceases.

In cells diffusion of many molecules including water occurs constantly. Water molecules diffuse in response to gradients in water potential. The rate of diffusion is given by Fick's first law:

$$J = \frac{\Delta\psi}{r}$$

Where: J = water flux density (g water m⁻² s⁻¹)

$\Delta\psi$ = difference in water potential.

r = resistance to diffusion.

$\Delta\psi$ is the water potential gradient, the driving force for diffusion. The rate of diffusion expressed as a flux is proportional to the magnitude of the concentration gradient.

3.4. Imbibition

Hydrophilic surfaces e.g. colloids such as proteins, starch and clay adsorb water. The tenacity with which the water molecules are adsorbed is a function of two factors namely:

- i) The nature of the surface,
- ii) The distance between the surface and the adsorbed molecules.

Those located directly on the adsorbing surface will be held extremely tightly. Those at some distance from the surface will be held less tightly.

The adsorption of water by hydrophilic surfaces is called hydration or imbibition. Forces of attraction either chemical or electrostatic are involved.

The matric potential (ψ_r) is a term used to account for all the forces causing imbibition i.e. holding water in a matrix (a material with surfaces that bind water).

The matric potential is a measure at atmospheric pressure of the tendency a matrix to adsorb additional water molecules. It is expressed in units of water potential e.g. a dry colloid or hydrophilic surface such as soil has an extremely negative matric potential, as low as – 300 Mpa.

*The same colloid in a large volume of pure water at atmospheric has a matric potential of zero i.e. it is at equilibrium with water.

The pressure generated by imbibition may be very great e.g imbibition pressure in a germinating seed splits the testa. A seed in a rock crevice may split the rock with the pressure of its imbibition water.

Imbibition of water by colloidal particles in cells helps them to withstand drought because of the tenacity with which imbibed water is held.

3.5. Osmosis.

The diffusion of water across a differentially permeable membrane from a region of higher potential (pure water) to one of lower potential (a more concentrated solution) is called osmosis.

A device that measures osmosis is called osmometer.

In it two things are present namely:

- i) Two volumes of solution or pure water are isolated from each other by a membrane that restricts the movement of solute particles more than it restricts the movement of solvent molecules.
- ii) There is a means of allowing pressure to build in one of the volumes.

A living cell is an osmosis system. In it the rigidity of the plant cell wall is responsible for the increase in pressure. The cell membrane allows water molecules to pass more rapidly than solute particles. It is the cell membrane that makes osmosis possible.

The potential with which pure water will diffuse toward a solution is the osmotic potential (ψ_{π}) of that solution. Osmotic potential (or solute potential) is caused by the presence of solute particles. Solutes decrease the potential of water since water diffuses from high potential (zero in pure water) to lower potential. The osmotic potential of a solution is always negative.

Water potential in cells:

If a cell is placed in a dilute solution or pure water whose osmotic potential is high (near zero) i.e. a hypotonic solution, water will diffuse in and the cell will be turgid.

If the cell is placed in a solution whose osmotic potential is equal to that of the cell sap (an isotonic solution), no net diffusion takes place and the cell will be flaccid i.e. it lacks turgor.

If the external solution is more concentrated than the cell sap (hypertonic solution), its osmotic potential is lower than that of the cell sap and water will diffuse out. The protoplasm will pull away from the cell wall as it shrinks and the cell will be plasmolysed.

LESSON 4 WATER RELATIONS

4.1 Introduction

In this lesson you will learn water relations of plant cells, soil moisture and water potential.

4.2. Objective

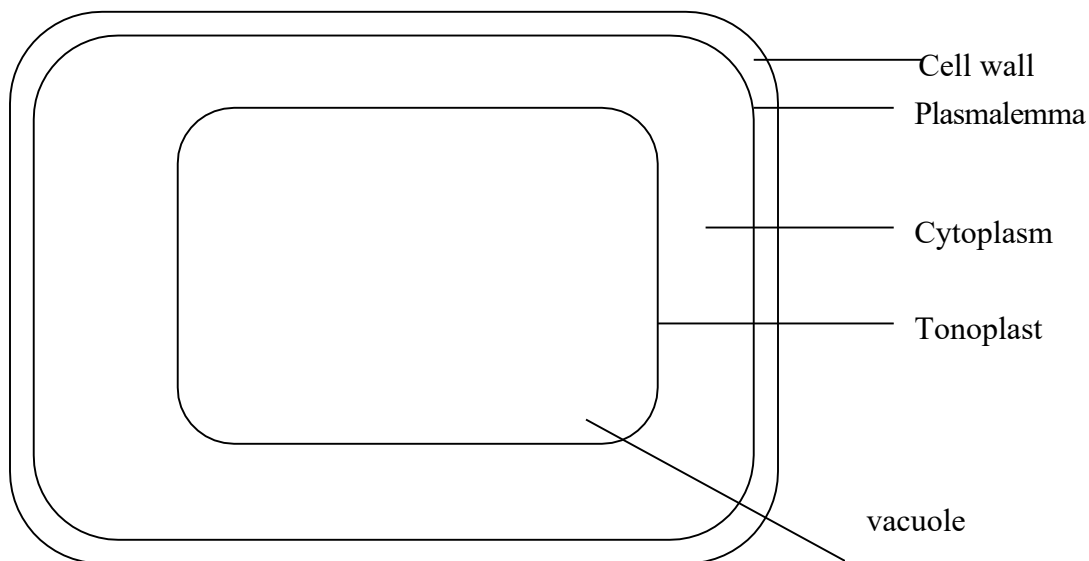
By the end of the lesson you should be able to appreciate:

- Cellular water relations
- Soil moisture
- Components of water potential
- Role of water potential gradient in water flow in the soil plant atmosphere system.

4.3. Content

4.3.1 Cellular water relation

- The cells involved in plant water relations are mature with a large fraction of the cell water contained in the vacuole.
- The cytoplasm, plasmalemma and tonoplast act as a complex differentially permeable membrane separating the vacuolar contents from the external medium.



- In a leaf, the external medium is the water in the cell walls and intercellular spaces (the leaf apoplast).
- Solute concentration in the apoplast is normally low. Water potential in the apoplast is therefore determined by the matric forces exerted by the cell walls (i.e. films of water are held by hydrophilic surfaces of cell walls. The matric potential (ψ_r) will be normally high, approximately -0.1 Mpa in a leaf well-supplied with water but not transpiring.
- In the vacuole matric forces are less important than in the apoplast. The vacuolar water potential is therefore determined by the solute concentration which may vary from -0.5 to 3.0 Mpa.

4.3.1.1. Water flux into the vacuole and leaf turgidity.

- Apoplastic water potential is higher (less negative) than vacuolar water potential. Since water flows from high to low potential. Water therefore flows across the cytoplasm into the vacuole.
- This raises the vacuolar water potential and increases the volume of the vacuole.
- Hydrastatic (turgor) pressure builds up within the vacuole.
- As turgor pressure rises adjacent cells press against each other. This results in the leaf becoming increasingly turgid.
- Equilibrium is reached when the difference in solute potential (driving force of water influx) equals the turgor pressure. Consequently the water potential in the cytoplasm is identical to that in the vacuole. Thus the cell water potential is as follows:

$$\Psi_{\text{cell}} = \Psi_{\text{cyt}} = \Psi_{\text{vac}}$$

4.3.2. Soil water

Water is present in the soil in various forms. It leaves the soil by:

- gravity
- evaporation
- absorption by plant roots.

When the soil is thoroughly wetted e.g. by a heavy rain and before water drains in response to gravity the soil is said to be saturated.

- much of this water moves down through pore spaces between soil particles and is referred to as gravitational water.

After gravitational water has drained out water that is left is held against gravity by adhesion between the water molecules and soil particles. This water is held in capillary pores and is referred to as capillary water.

Soil that contains are the capillary water it can held against gravity is said to be at field capacity (F.C.).

As water is removed from the soil it is held in ever increasingly thinner water films.

Finally the water potential in the soil and in the roots is the same and the roots is the same and the root cannot remove water from the soil.

Water however, continues to transpire from the shoots for a while so the plant wilts. Even if the plant is placed in a 100% relative humidity atmosphere, it cannot overcome the wilting.

At this point the amount of water in the soil to be at permanent wilting percentage (PWP).

- By convention, a soil with water potential of -1.5 Mpa is at PWP
- Amount of water at FC and PWP depends on:
 - Soil texture
 - Amount of soil organic matter.
- Water below the FC is held by attractive forces between the water molecules and the hydrophilic surfaces of clay particles and humus.

Soil water Potential

- Water uptake into roots takes place by osmosis. The osmotic potential of soil water and other forces by which water is held in the soil are key to understanding water movement in the soil.
- In the soil water diffuses from a region of high potential to one of low potential.

4.3.3. Components of water potential

- Pressure potential (P) which includes the operation of forces of gravity.
- Osmotic potential (Ψ_{π}) of the soil solution. (The potential with which pure water will diffuse towards a solution is the osmotic potential of that solution).
- The matric potential (Ψ_{τ}) a material with surfaces that bind water is called a matrix. The term matric potential is used to account for all the forces causing imbibition i.e. holding water in a matrix of any sort. It is an expression of chemical and physical attractions between water and soil particles.

The soil water potential is expressed as follows:

$$\Psi_{\text{soil}} = P + \Psi_{\pi} + \Psi_{\tau}$$

Where P = pressure potential

$$\Psi_{\pi} = \text{Osmotic potential}$$

$$\Psi_{\tau} = \text{matric potential.}$$

4.3.4. Water potential gradient in soil plant atmosphere system.

- Water moves in the soil plant atmosphere system in response to differences in the water potential in different parts of the system. Example, in a well watered plant which is transpiring, the water potential decreases progressively from the soil, via the xylem and the leaf apoplast to the bulk air. Water flows from the soil through the plant to the air in response to this gradient in water potential.

4.5. Questions

1. How does water flux from the external medium into the vacuoles of plant cells?
2. What is meant by the term permanent wilting percentage?
3. What are the components of soil water potential?

4.6 Activity.

Determination of the water content of a soil.

Weigh a small jam tin plus filter paper when dry. The tin should have small holes on the bottom side. Place some garden, black cotton or forest soil in the tin. Allow the tin to stand in shallow water in a pan or petri dish for 3 days. Place the can on a wire gauze to allow excess water to drain off for 48 hours. Then weigh the can and soil. Oven dry the can and soil at 105⁰C for 48 hours, cool and re-weigh. This should be repeated to reach constant weight. Work out the water content of the particular soil as a percentage of the dry weight of the soil.

References

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Delvin, R.M. and Witham, F.H. 1983. Plant Physiology Wordsworth Publishing company.

LESSON 5: MINERAL NUTRITION

5.1 Introduction

In this lesson you will learn the supply of nutrients to plants from the soil and plant growth as a function of nutrient concentration.

5.2. Objectives

By the end of the lesson you will be able to describe:

- effect of element concentration on growth rate.
- mineral uptake
- essential elements.

5.3 Content

5.3.1. Mineral nutrients in the soil.

Plants require a large number of elements which are either derived from minerals or mineralized during biological breakdown of organic matter. Mineral nutrients occur in the soil in both dissolved and bound form. Only about 0.2% of the nutrient supply is dissolved in soil water.

5.3.2. Essential elements.

There are two principal criteria by which an element can be judged essential or nonessential:

- An element is essential if a plant cannot complete its life cycle in the absence of that element.
 - An element is essential if it forms part of any molecule or constituent of the plant that is essential in the plant e.g. Mg in chlorophyll and N in proteins.
- Determination of essential elements to plants growing in a medium as complex as the soil is problematic. Plants are therefore grown with their roots in a solution of mineral salts referred to as nutrient solution. Growing of plants this way is called hydroponics.
 - Using this method Pfeffer, Sachs and Knops in 1860 showed that the following ten elements are essential: C, H, O, N, P, K, Ca, S, Mg and Fe.

- With the improvement of techniques to purify salts and water for hydroponics and to measure concentrations of elements, requirements of minute amounts of certain elements were observed. Thus, the list of essential elements includes also the following six trace elements: Mn, Zn, Bo, Cu, Mo, and Cl i.e. sixteen in all.

5.3.3. Micro- and Macro- nutrients

Of the sixteen essential elements seven namely, Mo, Cu, Zn, Mn, B, Fe and Cl are called trace elements or micronutrients because they are needed in tissue in small concentrations ie. equal to or less than $100 \mu\text{g g}^{-1}$ of dry matter.

The remaining nine S, P, Mg, Ca, K, N, O, C and H are the macronutrients i.e. they are needed in tissue concentrations of $1000 \mu\text{g}^{-1}$ dry matter or more.

5.3.4. Methods of studying plant nutrition.

i) Ash analysis

To detect some mineral elements, the plant is subjected to high temperatures (about 600°C) and the resulting ash analysed chemically. In the ash only mineral elements absorbed from the soil are present.

ii) Solution culture

Advantages:

- provides an excellent means for controlling the quantity and relative proportions of mineral salts given to a plant in an experiment.
- the excellent solvent characteristics of water.
- the ease with which water can be freed of contaminating influences

iii) Solid medium cultures.

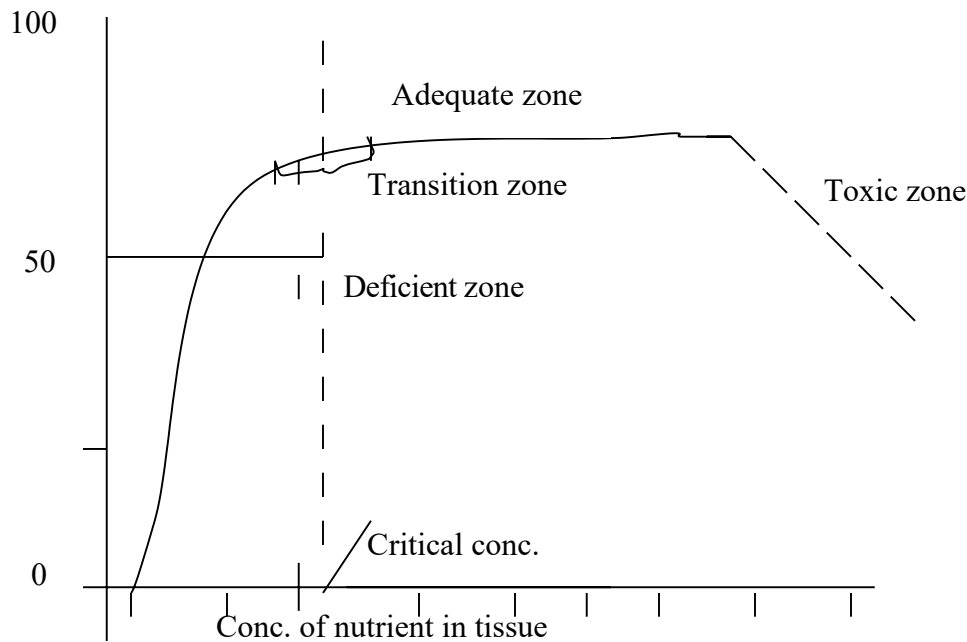
- Those used in plant nutrition studies are sand, crashed quartz and gravel.

Advantage of solid medium:

- Roots grow in a natural medium and no means of support needs to be provided.
- Nutrient solutions are added in three different ways:
 - pouring over the surface: slop culture.
 - dripping on the surface: drip culture.

- Forcing nutrient solution up from the bottom of the container- sub-irrigation. This requires a pumping apparatus.

5.3.5. Growth rate as a function of element concentration



Generalized plot of growth as a function of conc. of a nutrient in plant tissue.

- In the deficient zone growth increases dramatically as more of the element is provided and its concentration in the plant increases.
- Above the critical concentration i.e minimum tissue concentration giving almost maximal growth, increases in concentration do not appreciably affect growth. This is the adequate zone. It represents luxury consumption of the element.
- Continued increases of any element lead to toxicities or reduced growth (toxic zone).

*

5.3.6. Eutrophication

Excess nitrates and phosphates not absorbed by plants are leached from soils and appear in lakes and rivers where they cause the problem of eutrophication. This is nutrient enrichment leading to growth of algae and other aquatic plants. Upon death, the

decomposition of these plants by micro-organisms uses so much oxygen that fish and other animals die.

5.3.7. Nutrient deficiency symptoms

Plants respond to inadequate supply of an essential element by forming characteristic visually observable deficiency symptoms. These include:

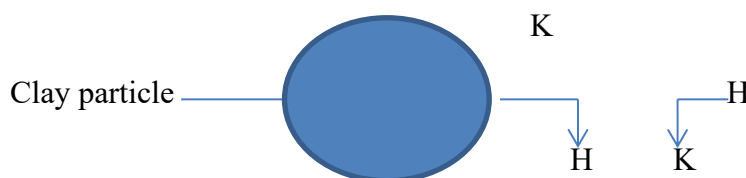
- Stunted growth of stems, leaves or roots
- Chlorosis i.e. lack of chlorophyll
- Necrosis i.e. death of various organs.
- The deficiency symptoms for any element depend on two factors;
 - i) Function(s) of that element.
 - ii) Whether or not the element is readily translocated from old to young leaves (nutrient mobility).

5.3.8. Uptake of mineral nutrients from the soil.

A root takes mineral nutrients from the soil in the following ways:

- i) Absorption of nutrient ions from the soil solution: these ions are available directly.
- ii) Exchange absorption of adsorbed nutrient ions: by releasing H^+ ions the root promotes ion exchange at the surface of clay and humus particles

obtaining in return nutrient ions. This process is referred to as cation exchange.



Cation exchange on a clay particle

Ion uptake into the cell.

Special properties of cell membranes assist in the entering of ions into the living cell e.g. proteins in membranes called carriers combine with the ions and transport them across the membrane.

Ion transport occurs down a concentration gradient (passive transport). Plants however also accumulate ions by transporting them against a concentration gradient (active transport). This requires a supply of energy. It therefore depends on the energy providing processes of respiration and photosynthesis.

5.4. Summary

In this lesson we looked at essential elements, micro and macro-nutrients and growth rate as a function of element concentration. We saw that an element is essential if a plant cannot complete its life cycle in the absence of the element. Microelements are required in trace quantities while macro-elements are required in higher quantities. We also saw that above the minimal concentration of an element giving almost maximal growth (critical concentration) increase in concentration does not increase growth appreciably.

5.5. Questions

1. What criteria are used to judge an element essential or non-essential?
2. What is the basis of classifying nutrients into micro and macro-nutrients?
3. What methods are used in studying plant nutrition?
4. What is meant by the term eutrophication?
5. How does element concentration affect plant growth?

5.6 Activity

Nitrates (in maize)

Cut out small sections of the stalk or leaf and place them in a petri dish or beaker. Apply a few drops of concentrated sulphuric acid and containing 1 per cent of diphenylamine. The older maize stalks may be cut lengthwise and the freshly exposed tissues tested for nitrates by applying a few drops of the test solution. The distribution of nitrates in the plant can be determined by testing the tissue of the stalk from the base to the top.

Nitrates are present in the tissue if a blue colour is produced immediately. This positive result is typical of all dark green maize plants. If no blue colour is produced, it shows that no reserve nitrates are present. This result confirms the yellowish green symptom indicating nitrogen deficiency.

***References.**

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LESSON 6: PHOTOSYNTHESIS

6.1 Introduction

In this lesson you will learn the two stages of the reactions of photosynthesis

6.2. Objectives

By the end of the lesson you should be able to:

distinguish the photosystems

describe non-cyclic and cyclic photophosphorylation.

6.3. Content

In eukaryotes, the reactions of photosynthesis take place in chloroplasts. There are two distinct stages:

- i) Light of certain wavelengths is trapped and converted (transduced) to chemical energy by a series of steps called light reactions.
- ii) CO_2 is reduced to organic compounds (sugars) by a series of steps called the dark reactions or CO_2 fixation.

Light absorption by chlorophyll:

Light comes in discrete packets of energy called photons or quanta. A pigment molecule can absorb only one photon at a time.

Electrons in molecules can exist in a number of discrete energy states or levels. The normal level called the ground state has the lowest energy.

Absorption of light by a chlorophyll molecule excites an electron raising it from the ground state to an excited state. The excited electron then returns to the ground state after transferring energy to a neighbouring molecule where another electron is excited. This mechanism of energy transfer is called resonance transfer. It is the mechanism by which energy harvested by many pigment molecules is funneled to one site.

The sites where the energy is funneled are the reaction centers where the excited electron is transferred from a pigment molecule to a non-pigment acceptor molecule. The high energy electron is then transferred to a series of other electron transport molecules.

Photosystems.

The photochemical phase of photosynthesis includes two distinct groups of functioning pigments referred to as photosystem I and photosystem II (PSI and PSII).

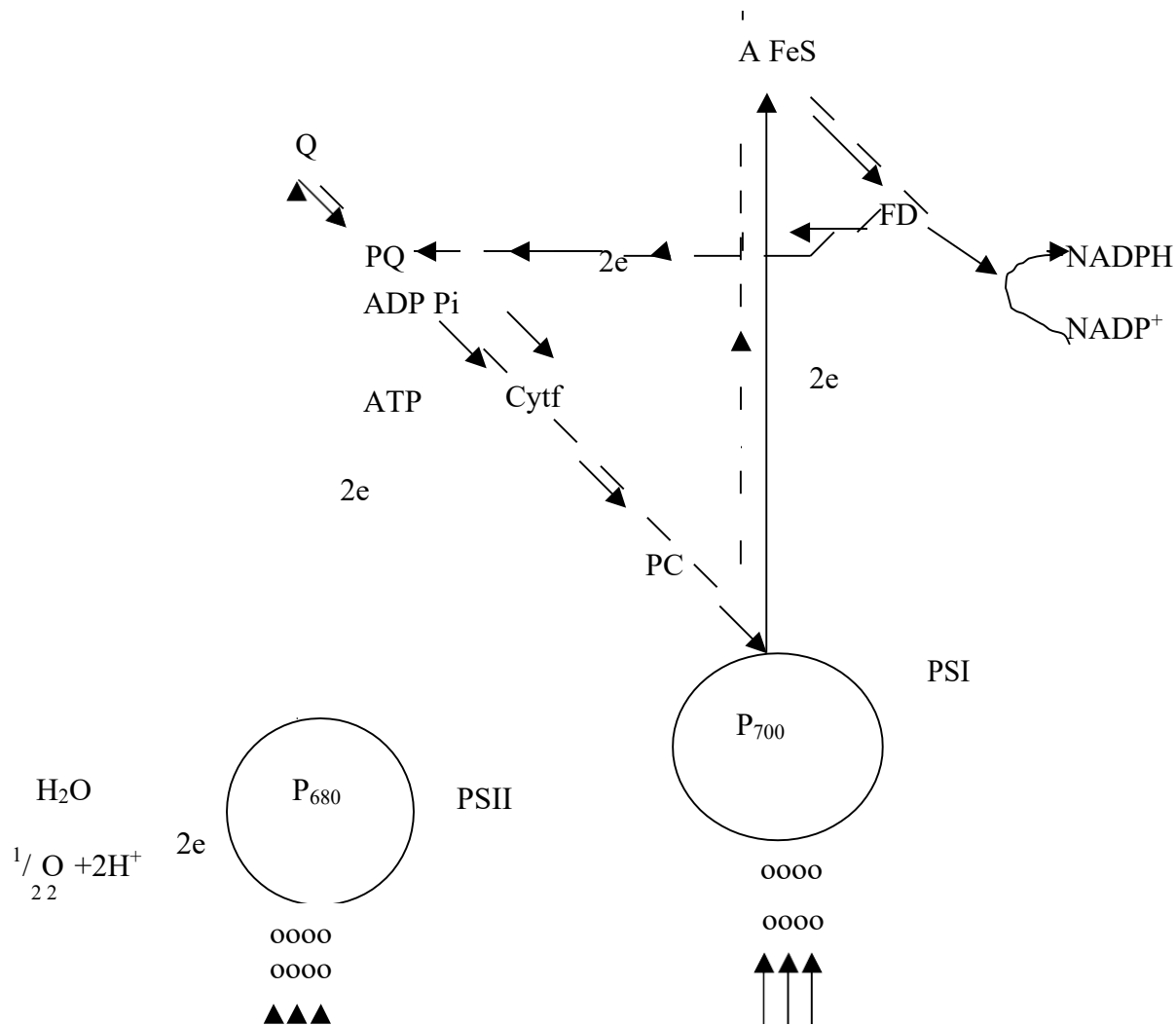
In both photosystems the pigments harvest light energy and transfer it by resonance to photochemically active centers called traps.

PSI: the active center pigment for PSI consists of chlorophyll a which absorbs light at 700 nm and is called P₇₀₀.

PSII: The chlorophyll a pigment at the center of PSII absorbs light at 680 nm and is called P₆₈₀

The Z-scheme and cyclic electron transfer.

The Z-scheme is so called because of its shape. It illustrates electron transport and the production of chemical energy in the form of ATP and NADPH in chloroplasts.



The Z- scheme and cyclic photosynthetic electron transport; -----> = cyclic electron transport. PQ = plastoquinone; PC= Plastocyanin; A(Fes) = iron-sulfur protein acceptor; FD = Ferredoxin.

Noncyclic photophosphorylation

The flow of electrons in both PSI and PSII is simultaneous through integrated reactions and photolysis of water. The integration of the two photosystems which is the means of ATP production in chloroplasts is called noncyclic photophosphorylation. Following excitation of P_{700} , electrons are passed to an iron-sulfur protein acceptor (AFe-S) and then to ferredoxin. Ultimately they are passed to $NADP^+$ resulting in the formation of NADPH.

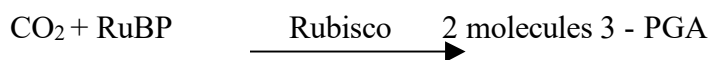
The transfer of electrons to $NADP^+$ creates an electron debit in PSI. This deficit is made up by excitation of P_{680} of PSII. The electrons are passed from P_{680} to P_{700} through a system of carriers namely Q (unidentified), Plastoquinone (PQ), cytochrome f (Cyt_f) and plastocyanin (PC). PQ passes electrons to Cyt_f at which point ATP is produced. The electron debit created in PSI is filled by electrons derived from photolysis (splitting) of water. The passage of electrons therefore requires both systems and results in the production of ATP and NADPH.

Cyclic Photophosphorylation

Activation of photosystem I causes electrons to flow from P_{700} to A(FeS). When electrons are not passed to $NADP^+$ they are passed to PQ and then back to P_{700} . The cycling of electrons from excited P_{700} and back as described is referred to as cyclic photophosphorylation. Chemical energy trapped in ATP and NADPH is used to power the fixation of CO_2 .

6.3.2 CO_2 fixation (Dark reactions)

CO_2 combines with a five-carbon sugarphosphate called ribulose –1, 5-bisphosphate (RuBP). This reaction yields two molecules of 3-phosphoglyceric acid (3-PGA), a three carbon acid. The reaction is catalysed by the enzyme ribulose bisphosphate carboxylase (Rubisco):



3 – PGA is used in a cyclic pathway to form sugar phosphates. The reactions of this pathway are collectively named the Calvin cycle or the C₃ pathway because the first product, 3 – PGA contains three carbons. The Calvin cycle occurs in the stroma of chloroplasts.

6.4. Summary

In this lesson we have looked at the two stages of photosynthesis in eukaryotes. We have seen there are two stages namely the light and dark reaction. In the light reactions the chemical energy required to power CO₂ fixation is produced. In the dark reactions CO₂ is fixed to produce a 3-carbon compound (3-Phosphoglyceric acid) that is used in the Calvin pathway to form sugar-phosphates.

6.5. Questions.

1. What do you understand by the term resonance transfer?
2. The photochemical phase of photosynthesis comprises two groups of functioning pigments. With the aid of a diagram describe them briefly.
3. Briefly describe the dark reactions of photosynthesis.

LESSON 7: RESPIRATION

7.1 Introduction:

In the previous lesson we learnt about the process of photosynthesis. During this process solar energy is trapped by green plants and transformed into chemical energy in the compounds produced. Solar energy is the ultimate source of energy that is available to living organisms. In this lesson we shall consider respiration which is a process in which the chemical energy is released and liberated. All living organisms require energy to drive the many activities that take place within the organism.

7.2 Objectives:

By the end of this topic the student should be able to:

- a) define and highlight the importance of respiration.
- b) distinguish between respiration and photosynthesis.
- c) identify substrates by their respiratory quotients.
- d) state the reaction sequences of glycolysis.
- e) trace the fate of pyruvic acid in respiration.
- f) discuss the factors that affect respiration.

7.3 Content

7.3.1 Definition and significance

Respiration is a chemical process in which complex carbon containing organic compounds are oxidized and decomposed into simpler substances with the release of energy. The carbon containing compounds are glucose and other organic compounds. The reaction is oxidative and so there is an intake of oxygen. The simpler substances liberated are carbon dioxide and water. Potential energy is converted into kinetic energy. The energy is released in usable forms. The reaction is exothermic because part of the energy released is in the form of heat.

The essential features of respiration are: absorption of oxygen, oxidation of food (conversion of carbohydrate to carbon dioxide and water); release of energy part of which is used to drive vital process and the rest lost as heat; formation of intermediate products having different roles in metabolism and loss of weight in plants as a result of oxidation.

The overall process of respiration is represented by the equation:



Respiration is significant because

- i) it releases energy which is used in various metabolic processes that are essential for plant life and activates cell division.
- ii) compounds that are necessary in participating as important cell constituents are formed.
- iii) insoluble food is converted to soluble form.
- iv) carbon dioxide liberated partly plays an active role in maintaining the balance of carbon cycle in nature.
- v) stored energy (potential energy) is converted to usable form (kinetic energy).

Respiration is a reverse of photosynthesis. Differences between these two important processes are given in the table below:

Feature	Respiration	Photosynthesis
1. Oxygen	Absorbed	Liberated
2. Carbon dioxide	Evolved when carbon containing compounds are oxidized	Absorbed and fixed to form carbon containing compounds
3. Occurrence	Occurs during day and at night	Occurs in presence of light (during the day)
4. Light	Not essential	Essential
5. Energy	Potential energy is converted to kinetic energy	Light (radiant) energy is converted to potential energy (chemical energy)
6. Raw materials	Glucose and oxygen	Carbon dioxide and water.

7. Chlorophyll	Not necessary	Necessary.
8. Energy flow	Energy is released (Exothermic reaction)	Energy is absorbed (Endothermic reaction)
9. Weight	Lost	Gained
10. Food	Stored food destroyed (catabolic process)	Food manufactured (Anabolic process)
11. Processes	Involves dehydrolysis and decarboxylation	Involves hydrolysis and carboxylation
12. ATP	36 ATP molecules are formed per glucose molecule	18 ATP molecules are required to synthesize a glucose molecule.

7.3.2 Respiratory Quotient.

Organic substances are substrates in respiration. These substrates are oxidized and carbon dioxide is released. The gas exchange (intake of oxygen and release of carbon dioxide) can be used to identify the substrate being oxidized. A ratio known as respiratory quotient (RQ) is used.

Respiratory Quotient is the ratio of the volume of carbon evolved and oxygen taken in simultaneously by a given weight of tissue in a given period of time at standard temperature and pressure.

$$\text{Respiratory Quotient (RQ)} = \frac{\text{Volume of CO}_2 \text{ evolved}}{\text{Volume of O}_2 \text{ absorbed.}}$$

The respiratory quotients of some respiratory substrates are given in the table below.

Substrate	RQ
Carbohydrates	1.00
Fats	0.70
Proteins	0.80
Organic acids	>1.00
i) Malic acid	1.33
ii) Tartaric acid	1.60
iii) Citric acid	1.33
iv) Oxalic acid	4.00

Carbohydrates are the principle respiratory substrates. Common examples are starch, insulin, sucrose, glucose and fructose. Complex carbohydrates are first hydrolysed to simple sugars or hexoses. Hexoses give equal volumes of CO₂ evolved and O₂ consumed. The RQ is therefore unity.

Fats are poorer in oxygen. They require more oxygen for complete oxidation. They are first hydrolysed to glycerol and fatty acid. The RQ is less than unity.

Proteins and amino acid derivatives are substrates only during starving conditions. They have less oxygen therefore they require more oxygen for complete oxidation. Respiratory quotient is also less than unity.

Organic acids are rich in oxygen and the proportion of oxygen to carbon is very high. Less oxygen is absorbed and more CO₂ is evolved making the RQ to be more than unity.

7.3.3 Glycolysis.

Respiration can be divided into two distinct phases: glycolysis and aerobic or anaerobic respiration. Glycolysis is also referred to as Embden Meyerhof – Parnas (EMP) pathway after people who traced it. In this glycolytic pathway every glucose molecule is oxidized to two molecules of pyruvic acid (a three carbon organic acid). In plants D-glucose and

D-fructose are the main monosaccharides catabolised by glycolysis. It takes place in the cytoplasm and produces key metabolic components of the respiration process.

There are three distinct steps in the process of glycolysis: phosphorylation, sugar cleavage and pyruvic acid formation.

i) Phosphorylation:

In this step the sugar is prepared for reaction by the addition of phosphorus. Sugar cannot combine with phosphates directly.

The process is controlled by enzymes. It is an energy requiring process. One phosphate group is transferred at a time to the sugar molecule that becomes phosphorylated. Two ATP molecules are involved and two phosphate group are added to the sugar, each at each end of the sugar molecule giving the compound fructose -1, 6 -diphosphate.

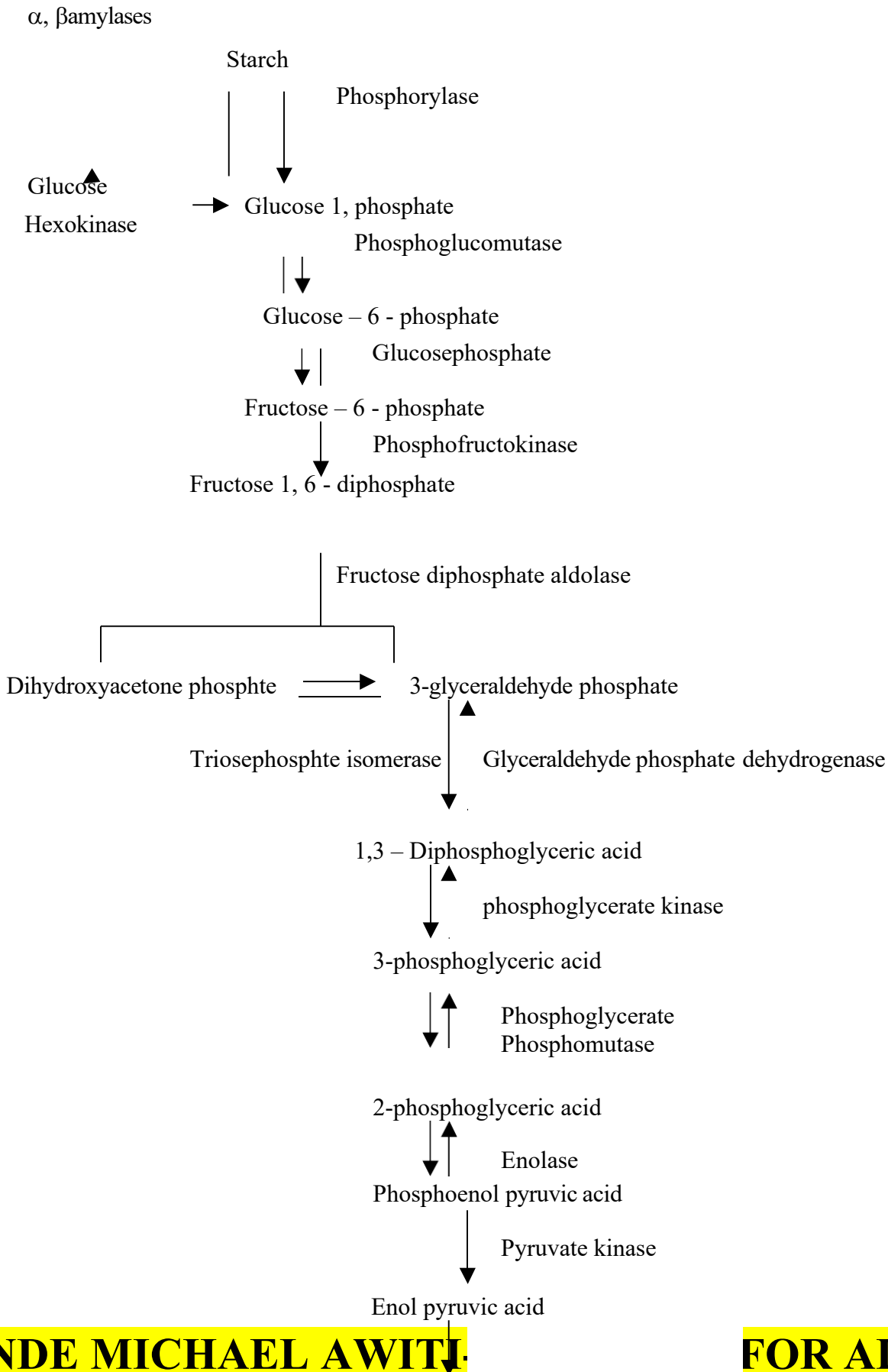
ii) Sugar cleavage

The sugar to be split is fructose 1, 6 diphosphate. Through catalysis by an enzyme the sugar is split into two different three carbon atom sugars (trioses). The two fragments are dihydroxyacetone phosphate and D-glyceraldehyde 3- phosphate (3-phosphoglyceraldehyde).

iii) Pyruvic acid formation.

The two trioses can be interconverted to one another and are in equilibrium. For glycolysis to proceed the equilibrium tilts in favour of formation of more D-glyceraldehyde 3-phosphate. This is then used in the formation of pyruvic acid through a number of transformations.

The whole glycolytic pathway reactions are indicated as follows:



OTONDE MICHAEL AWITI-DELEGATE FOR ALL

Spontaneous reaction

Pyruvic acid

OTONDE MICHAEL AWITI-DELEGATE FOR ALL

Once formed pyruvic acid enters the next phase of respiration. The fate of pyruvic acid depends on the presence or absence of molecular oxygen which results in aerobic respiration. The absence of molecular oxygen gives anaerobic respiration.

(ref. Section 16.1 Glycolysis Is an Energy-Conversion Pathway in Many Organisms. Biochemistry. 5th edition. Berg JM, Tymoczko JL, Stryer L. New York: W H Freeman; 2002).

7.3.4 Aerobic respiration.

Pyruvic acid is oxidized stepwise to CO_2 and H_2O with the release of all available energy that it contains. Only about 20% of the available energy in glucose molecule is released during glycolysis. Pyruvic acid contains about 80% of the energy locked up in it i.e the two pyruvic acid molecules produced. During the oxidation of pyruvic acid between 40% and 50% of the energy is released and trapped in a potentially useful form in ATP or reduced nucleotide. The remaining amount is lost in form of heat.

Aerobic respiration can be divided into three main steps:

- i) The entrance of carbon of pyruvic acid into the organic cycle of respiration.

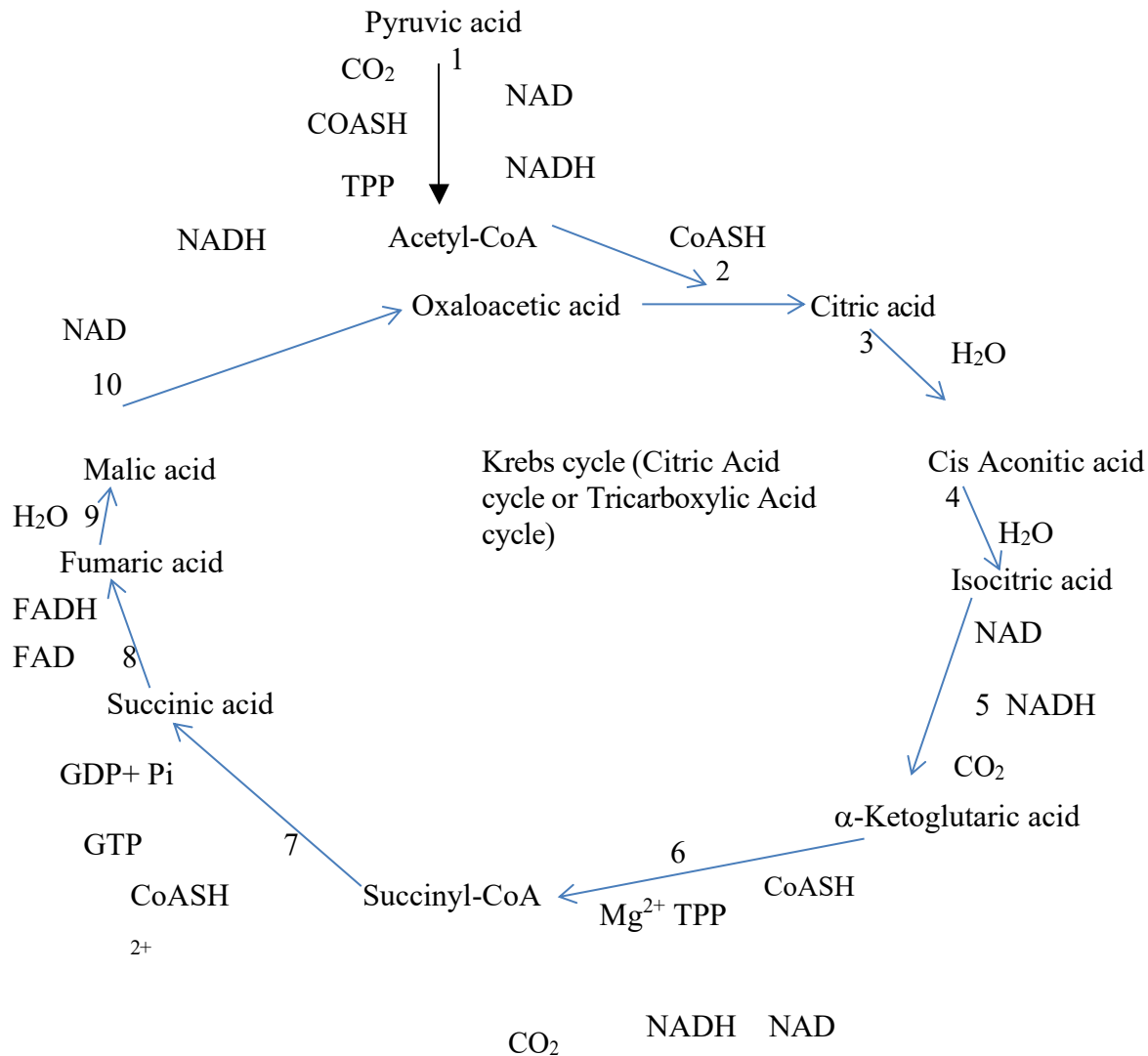
Pyruvic acid is oxidized and a molecule of CO_2 is formed, two electrons, two hydrogen atoms and some energy is transferred to NAD which is converted to NADH. What remains of pyruvic acid is a two-carbon compound. This compound actually enters the organic cycle when acetyl-CoA donates acetyl group to the acceptor molecule, a four-carbon atom oxaloacetic acid. As a result a six-carbon atom, citric acid is formed.

- ii) The organic acid cycle of respiration.

Citric acid is gradually broken down in a cyclic series of reactions involving seven other organic acids including oxaloacetic acid. This cycle of reactions is commonly referred to as the Krebs cycle after H.A Krebs (1937) who proposed the cycle of reactions in order to explain the fate of pyruvic acid in the presence of oxygen. The cycle is also known as Citric Acid cycle because of citric acid which is an important intermediate of the cycle. Tricarboxylic Acid cycle (TCA) is also used because of citric acid and isocitric acid

which are important intermediates of the cycle. Tricarboxylic Acid cycle (TCA) is also used because of citric acid and isocitric acid that have three carboxyl groups (COOH).

In these reactions pairs of electrons (and hydrogen atoms) are transferred to electron carriers, the coenzymes NAD or NADP and FAD (Flavin Adenine Dinucleotide). Two molecules of CO₂ are produced and a new oxaloacetic acid capable of accepting another acetyl group is formed. The cycle which takes place in the mitochondria is given below:



GTP - Guanosine Triphosphate

TPP - Thiamine pyrophosphate

SH - Sulfhydryl group

FAD - Flavine Adenine Dinucleotide

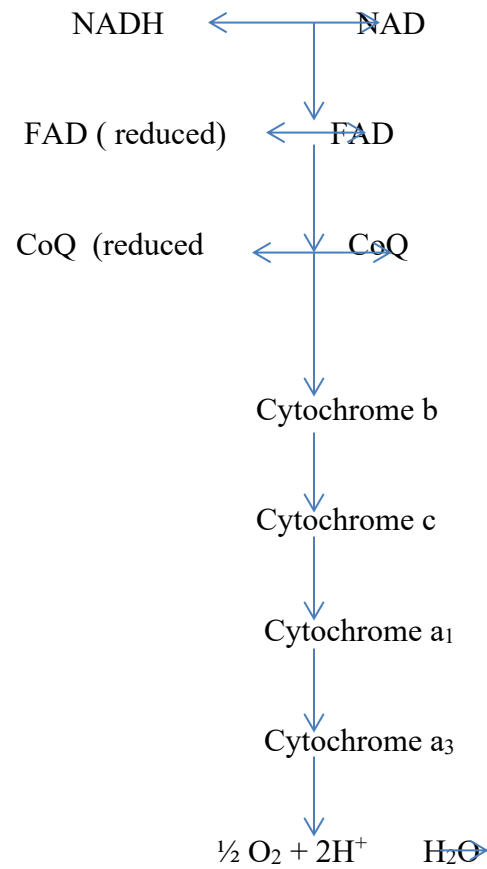
The reactions are catalyzed by several enzymes given in the diagram as (1-10)

1. Pyruvic dehydrogenase
2. Citric synthase
3. Aconitase
4. Aconitase
5. Isocitric dehydrogenase
6. α -ketoglutaric dehydrogenase complex
7. Succinic thiokinase
8. Succinic dehydrogenase
9. Fumarase
10. Malic dehydrogenase.

iii) The Electron Transport System

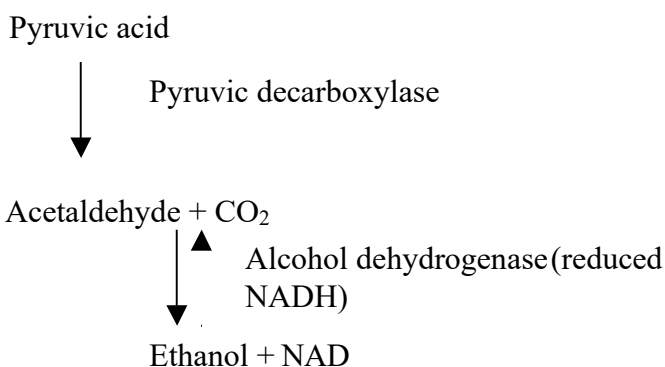
Nucleotides NADH and FADH are produced in the Krebs cycle. These reduced nucleotides are oxidized stepwise in the final series of reactions called electron transport system. The energy present in the nucleotides is not used directly by the cell but is temporarily stored in ATP. Pairs of electrons are transferred from the reduced coenzymes through several electron carriers. These carriers include Coenzyme Q (CoQ) and the cytochromes b,c,a, and a_3 .

A pair of electrons is eventually passed on to oxygen which is the terminal electron acceptor. Two hydrogen ions from the cellular environment combine with oxygen to form water. As the electron transport takes place three molecules of ATP are synthesized for each NADH (or NADPH) oxidized. Two molecules of ATP are synthesized for each FADH oxidized. This ATP formation which is linked to oxidation reactions of respiration (electron transport) is known as oxidative phosphorylation.



7.3.5 Anaerobic Respiration

In the absence of oxygen pyruvic acid is converted to alcohol (Ethanol) and CO_2 . The process is also called alcohol fermentation. It takes places in two steps: Pyruvic acid is converted to acetaldehyde and CO_2 . Acetaldehyde is then converted to ethanol. The respective enzymes are pyruvic decarboxylase and alcohol dehydrogenase.



In some organisms e.g. lactic acid bacteria anaerobic respiration results in the formation of lactic acid instead of alcohol. Acetaldehyde is converted to lactic acid through the activity of lactic acid dehydrogenase. Other organic acids such as acetic and succinic acids can also be produced in anaerobic respiration.

7.3.6 Differences between aerobic and anaerobic respiration.

The differences between aerobic and anaerobic respiration are given in the table below

Feature	Aerobic	Anaerobic
Oxygen	Present	Absent
Ethanol	Not produced	Produced
Lactic acid	Not produced	Produced
ATP	Produced	Not produced
H_2O	Produced	Not produced
Energy	Most is released	Most unreleased
Enzymes	Several	Two enzymes
Coenzymes	NAD, FAD	Only NAD
Electron transport	Elaborate	Limited

Microorganisms	Absent	Present
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7.3.7 Factors affecting the rate of respiration.

a) External factors

1. Temperature

At extreme temperatures, the rate of respiration declines. At very low temperatures the rate of respiration is low. This is the reason why fruits and vegetables are stored at low temperatures. The food material or respiratory substrate is not degraded much. It also explains why potato tubers grown on the hills are larger in size. The respiration rate is slow and more food material is stored and tuber size increases.

At high temperatures the rate of respiration decreases because enzymes are denatured, oxygen entry into cells may not be fast enough, removal of CO₂ from cells is not fast enough and accumulates and that respiratory substrate becomes limiting.

2. Light.

The effects of light on respiration are indirect. The rate of respiration increases in light because:

- i) photosynthesis takes place in light providing more respiratory material.
- ii) increases the temperature which causes an increase in respiration
- iii) it affects the opening of the stomata which regulates gas exchange.

3. Oxygen concentration

Oxygen is a raw material in respiration. Reduction of oxygen concentration reduces the rate of respiration. It reaches the minimum at 1% concentration and below this CO₂ concentration rises.

4. Carbon dioxide concentration

This has an inverse relationship with the rate of respiration. If the concentration of CO₂ increases the rate of respiration falls. This is why seeds fail to germinate in an atmosphere

of carbon dioxide and fruits and vegetables can be preserved in an atmosphere of carbon dioxide.

5. Water.

With adequate amounts of water the rate of respiration is not affected. Shortage of water reduces the rate of respiration.

6. Certain chemical substances:

- i) inorganic salts – it has been observed that the rate of respiration increases when a plant tissue is placed in a salt solution.
- ii) enzymatic inhibitors – some inhibitors such as cyanides, azides, carbon monoxide, iodoacetate in very small quantities increases the rate of respiration. The rate falls as the concentration increases.
- iii) anaesthetics – chloroform, ether, acetone, formaldehyde, alkaloids and glucosides act in the same manner as enzymatic inhibitors.

7. Mechanical stimulation.

Mechanical treatments such as handling, stroking or bending increases the rate of respiration. Increases of between 20% and 183% have been reported in leaf respiration.

8. Wounding or injury

Rate of respiration increases following wounding or injury. This is because of increase in respiratory substrates such as sugar.

b) Internal factors.

1. Protoplasmic factors.

The rate of respiration is affected by the quantity and quality of the protoplasm. Younger cells have more protoplasm and respire more rapidly. Older cells have lesser protoplasm because of larger vacuoles and much wall materials. Ageing reduces rate of respiration. Qualities of enzymes also affect rate of respiration.

2. Respiratory material

Rate of respiration increases in the presence of respiratory material.

7.4 Summary.

Respiration is an important process as far as provision of energy is concerned. The energy is required to drive the various activities taking place in the cell. Various substrates are converted in respiration through glycolysis and conversion of pyruvic acid to CO₂, water, ethanol organic acids and energy. Several external and internal factors enhances or reduces the rate of respiration.

7.5 Activity.

Release of heat during respiration

Place the same amount of ungerminated seeds in two thermos flasks with little moisture. The flasks are closed with airtight corks. A thermometer is placed in each flask through a hole in the cork ensuring that it is also airtight. Temperature is recorded before and after some time and Compare the two temperatures.

7.6 Questions.

1. Why is respiration important to plants?
2. State the essential features of respiration.
3. What are the distinguishing features of respiration and photosynthesis?
4. Briefly discuss the respiratory quotients of respiratory substrates.
5. Briefly describe the reaction sequence of glycolysis.
6. Discuss the fate of pyruvic acid in aerobic respiration.
7. Explain why fruits and vegetables can be preserved in an atmosphere of carbon dioxide.
8. Explain why potato tubers grown on the hills are larger in size.

LESSON 8: CARBOHYDRATES, LIPIDS, AMINO ACIDS AND PROTEINS.

8.1 Introduction:

Carbohydrates, lipids, amino acids and proteins are primary plant products. They are found in all plant species and distributed within the whole plant. They are all biologically important biomolecules. The primary elements of carbon, hydrogen and oxygen are found in all these compounds. In addition nitrogen and sometimes sulfur are found in amino acids and proteins. During starvation lipids, amino acids and proteins are converted to carbohydrates, which are used as a source of energy. In this lesson we look at the classification, structures and basic characteristics of these compounds.

8.2 Objectives

By the end of this topic, students should be able to

- a) define carbohydrates, lipids, amino acids and proteins.
- b) classify these compounds into various groups.
- c) identify the specific properties of the biomolecules.
- d) write their structures and identify the functional groups

8.3 Content

8.3.1 Carbohydrates

a) Definition

Carbohydrates are a group of organic compounds that contain carbon, hydrogen and oxygen, generally in the ratio of 1:2:1 (CH_2O). They are then considered as hydrates of carbon. The definition has been broadened to include compounds that contain nitrogen, sulfur and compounds which do not confirm to the ratio of 1:2:1. Carbohydrates have also been defined as polyhydroxyaldehydes, polyhydroxyketones and their derivatives.

b) Importance

Carbohydrates are significant to plants in three ways:

- i) They are means of storage of energy. The energy ultimately comes from the sun as light energy. Through the process of photosynthesis light energy is converted to chemical energy. This conversion is called transduction.
- ii) They are constituents of supporting tissues which are basic to plant form e.g erect growth.
- iii) They provide carbon skeletons that are used in the synthesis of other organic compounds.

c) Classification

Carbohydrates are broadly classified into monosaccharides, oligosaccharides and polysaccharides.

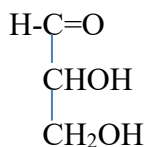
Monosaccharides:

These are the least complex of the carbohydrates. They do not yield any simpler carbohydrates when hydrolysed. They are the building blocks of more complex carbohydrates. Acetic acid and formaldehyde are carbohydrates only according to their definition as hydrates of carbon but lack other physical and chemical properties of carbohydrates.

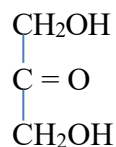
Monosaccharides are classified further according to the number of carbon atoms:

(i) Trioses

They contain three carbon atoms in their structures. Examples are glyceraldehydes and dihydroxyacetone.



Glyceraldehyde

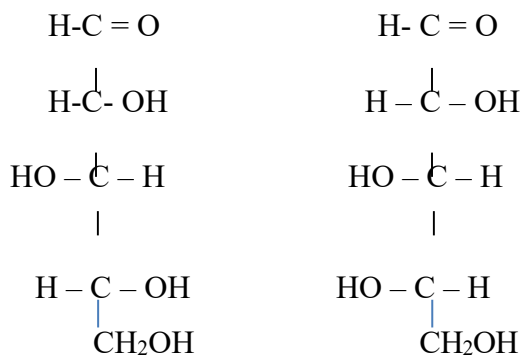


Dihydroxyacetone

These two examples are reducing sugars because they have reducing groups (-CHO) for glyceraldehyde and (-CO) for dihydroxyacetone on carbon 1 and 2 respectively. Sugars can be divided into aldoses (-CHO) and ketoses (-CO) according to the reducing groups.

ii) Pentoses

These are five carbon sugars. They are mainly found in bound form, as constituents of some complex carbohydrates. They are rarely found dissolved in the cytoplasm (in free state). Examples are: D-xylose, L-arabinose, D-ribose and 2-deoxy-D-ribose.



D-xylose (found in xylans)

L-Arabinose (found in Arabans)

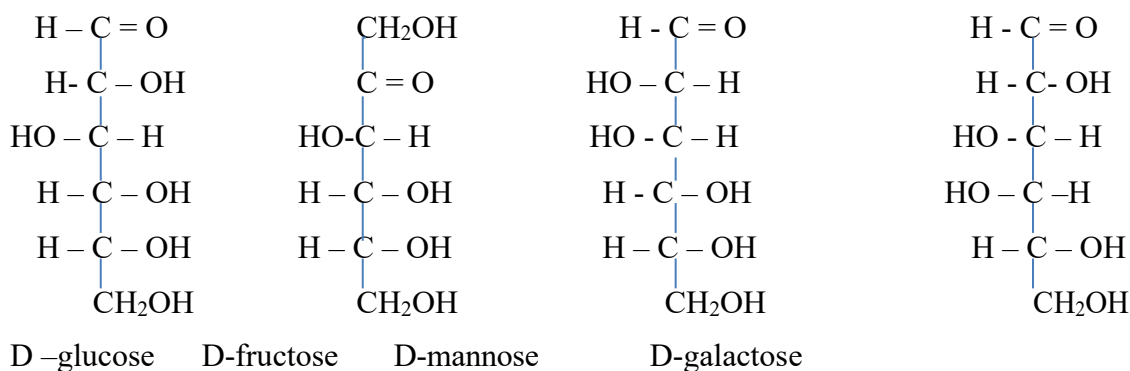


D – ribose (found in RNA)

2-deoxy- D-ribose (found in DNA)

iii) Hexoses;

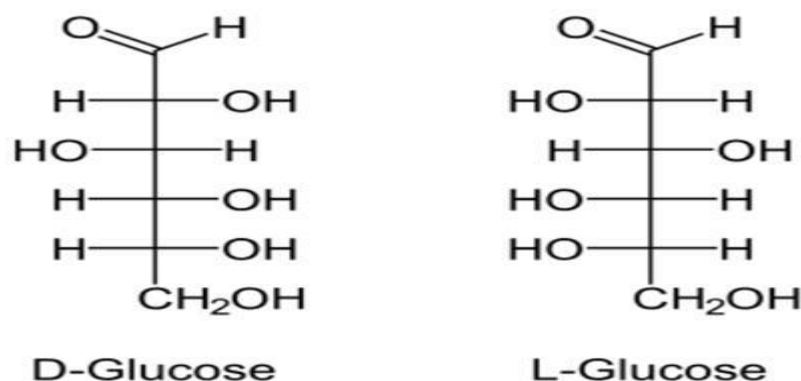
These are six carbon sugars found in a free state in the cell or as components of complex carbohydrates. Common examples are D-glucose, D-fructose, D-mannose and D- galactose. The only ones found dissolved in free state are D-glucose and D-fructose.



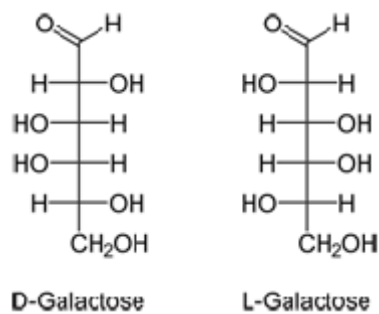
All these hexoses are asymmetric in that they have four different substituents. Their arrangement gives rise to several diastereoisomers. They differ in physical, chemical and biological properties.

Sugars can be divided into two using the property of optical rotation. The two types or mirror compounds are called enantiomers. They are identical in all physical properties except optical rotation. Optical rotation is a property in which a plane of polarized light will be rotated either to the left (levorotatory) or right (dextrorotatory) when light is transmitted through the mirror compounds. The sugar that deflects light to the left is prefixed l(-) and to right d(+) e.g l(-) glucose and d(±) glucose.

Another division uses the **configuration of the highest center of asymmetry which is carbon 5 for glucose**. This gives D-series and L-series if the hydroxyl group is to the right and left respectively e.g D-glucose and L-glucose.



D configuration is shown by all sugars found in plants. L-galactose is natural and occurs as a constituent of agar.

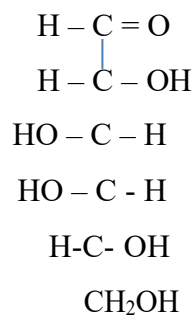


d) Carbohydrate structures:

Structures found in carbohydrates are straight chain, branched chain and ring structures.

i) Straight chain structure

This is shown by glucose. It has a straight carbon chain.

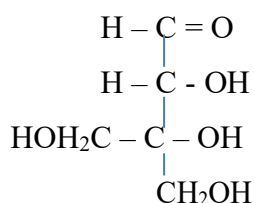


Glucose

ii) Branched chain structure

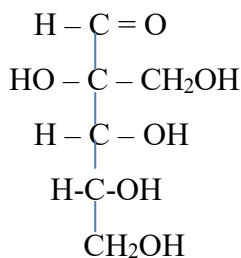
This is shown by two monosaccharides: Apinose and Hamamelose.

Apinose is a five carbon sugar. It was first identified from Parsley and Arrowwood plants. It is widespread and in some cases found in large quantities. Carbon chain is branched.



Apinose

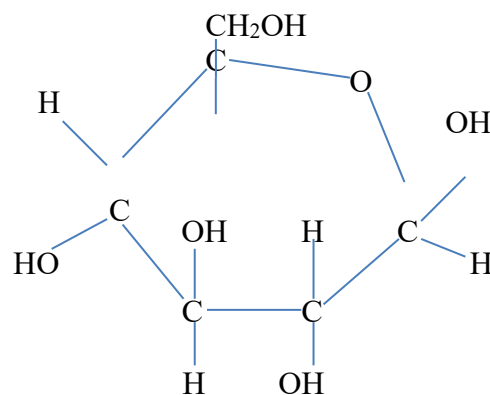
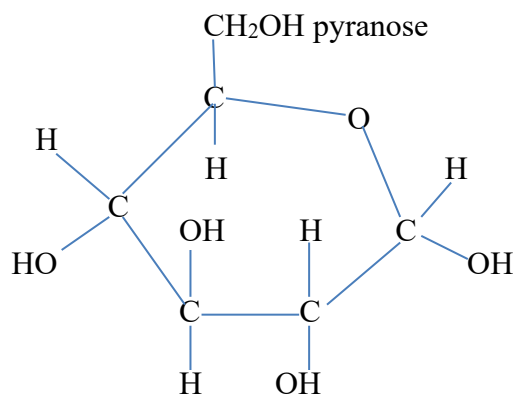
Hamamelose is a six carbon sugar. It was first discovered in the bark of Witch hazel. It is of widespread occurrence in higher plants especially primula species.



Hamamelose

iii) Ring structure.

It is a feature of cyclic and ring forms of carbohydrates. For example, when glucose is in solution carbon 1 and carbon 5 come into close proximity. A bond called oxygen bridge is established between the two carbons resulting in ring structure. A new center of asymmetry is created around carbon 1. The position of the hydroxyl on this carbon is used to divide sugars into α - and β -forms. In the case of glucose there are α -D-glucose with OH- below and β -D-glucose above the ring.



e) Oligosaccharides

Oligos is a Greek word meaning “few”. Oligosaccharides are carbohydrates that yield generally 2 to 6 molecules of simple sugars. They are usually crystalline, soluble in H₂O and have a sweet taste. The number of monosaccharide units has been used to classify them into the various oligosaccharides:

Disaccharide – two monosaccharide units

Trisaccharide – three monosaccharide units

Tetrasaccharide – four monosaccharide units

Etc.

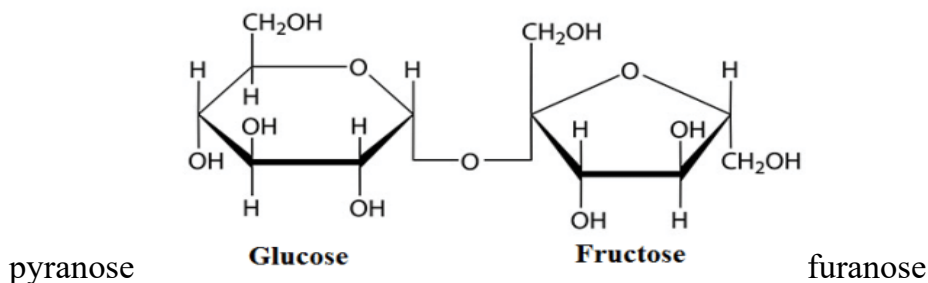
Disaccharides:

Examples are sucrose, maltose, cellobiose and lactose.

i) Sucrose:

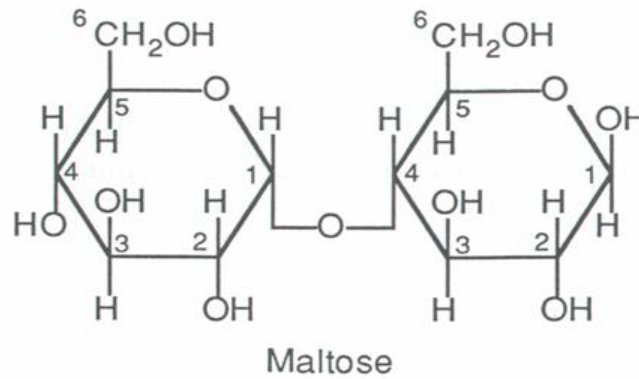
Is the principle disaccharide of higher plants. It is a condensation product of glucose and fructose. Some plants such as sugarcane and sugar beet produce it in large quantities. Though it is formed from reducing sugars, glucose and fructose, sucrose is a **non-reducing sugar**. This is so because the reducing groups of glucose and fructose have been involved in the formation of an oxygen bridge between glucose and fructose. It is formed between carbon 1 of glucose and carbon 2 of fructose. This type of bond (linkage or link) is α -1,2. Glucose occurs as a six-membered ring called pyranose ring whereas fructose occurs as a five-membered ring called furanose.

Sucrose is the principal form in which carbohydrates are transported in higher plants.



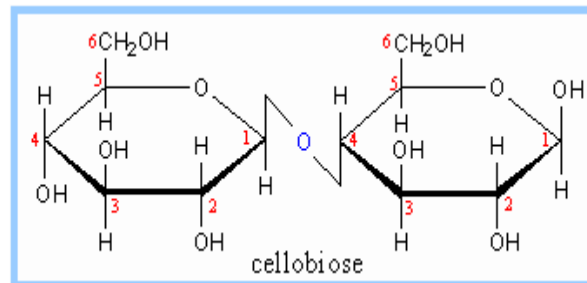
ii) Maltose

Is a product of partial degradation of the polysaccharide, starch. It is composed of two glucose molecules held together by a α -1,4 linkage. It is a reducing sugar because there is a free aldehyde group that is responsible for the reducing property.



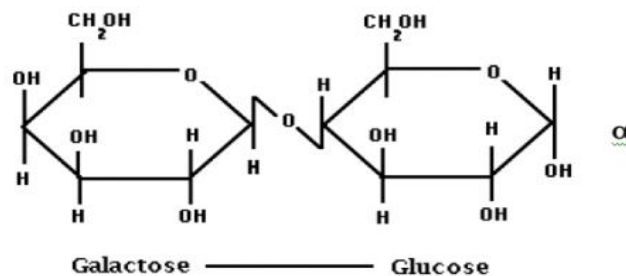
iii) Cellobiose

This is a product of partial degradation of cellulose or lignin. It is composed of two glucose molecules that are held together by β -1,4 linkage. It is also a reducing sugar.



iv) Lactose:

This is synthesized in the mammary tissue and is the main sugar in milk. It has however been detected in minute quantities in flowers of some plants. It is a disaccharide of glucose and galactose joined by β -1,4 linkage. (Omit)



Trisaccharides:

Raffinose

This trisaccharide occurs naturally in many plants. It is found in large quantities in storage organs eg. seeds. Small amounts have been reported in leaves of many plants.

The monosaccharides forming it are glucose, fructose and galactose. It is a non-reducing sugar.

i) Gentianose:

It occurs naturally in many plants. It consists of two molecules of glucose and one molecule of fructose. It is a non-reducing sugar.

Tetrasaccharides:

These are oligosaccharides that contain four monosaccharide units. An example is stachyose which is found in many tree species. It is formed from glucose, fructose and two molecules of galactose. It is the principal carbohydrate transported in some plants e.g the pumpkin (*Cucurbita pepo*).

f) Polysaccharides

These are compounds that generally have very high molecular weights. They differ from monosaccharides and oligosaccharides by being insoluble in water and lacking in sweetness. Polysaccharides are formed from many repeating units of monosaccharides called monomers. They are in turn referred to as polymers. The commonest example are starch and cellulose.

i) Starch

It has high molecular weight. On complete hydrolysis it yields α -D-glucose molecules. These molecules are produced in photosynthesis and converted to starch grains. The grains are prevalent in storage organs such as seeds, tubers and bulbs. They are reserve nutrients to be used for new growth and development arising from the storage organs.

Generally starch is considered as a straight chain polymer that consist of amylose and amylopectin. These are large molecules that yield α -D-glucose units on hydrolysis.

Amylose is a straight chain polymer. It has only α - 1, 4 linkages joining the glucose units. It is soluble in water. The blue-black coloration obtained when iodine is added to starch is due to amylose.

Amylopectin is a branched chain molecule. It has α - 1, 4 linkages, α - 1,6 linkages and sometimes α - 1,3 linkages. Since it has a complex structure, amylopectin is less soluble in water. It gives red to purple coloration when starch reacts with iodine.

ii) Cellulose

Cellulose is a straight chain polymer with a high molecular weight. It consists of glucose units that are linked together by β - 1,4 linkages. It is a basic component of the cell wall and is the most abundant natural product in the world. The percentage of cellulose in cell walls ranges from approximately 20% in the primary cell and 43% in the secondary cell walls. Cellulose is a relatively inert material but can only be degraded by strong chemical treatments eg. conc. HCl, H₂SO₄ or NaOH. Glucose molecules are obtained when these treatments degrade cellulose. It is insoluble in water but can be dissolved in ammoniacal solutions of cupric (Copper II) salts.

Cellulose is of no nutritive value to humans because of lack of cellulase enzymes.

Ruminants, termites, bacteria and some protozoa can digest cellulose. In plants cellulose is useful for structural purposes.

iii) Pectic substances.

These are a group of closely related polysaccharides. There are three groups of pectic compounds found in plants: pectic acid, pectins and protopectins. Pectins and protopectins are derivatives of pectic acid.

Pectic substances are found most abundantly in the middle lamella between cell walls.

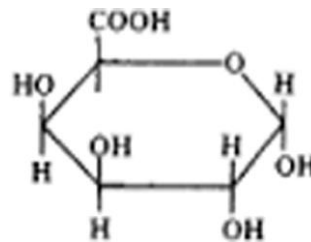
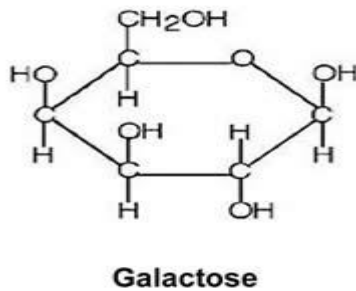
They occur mostly as calcium and magnesium salts of pectic acid (pectates).

a) Pectic acid

It is an unbranched molecule that consists of about 100 D-galacturonic acid residues.

These are linked by α - 1,4 linkages. A galacturonic acid molecule is formed from

galactose when CH_2OH in position 6 is oxidized to carboxyl group COOH .



Galacturonic acid

Pectic acid is insoluble in water and may be precipitated by calcium ions.

b) Pectins

These are short chains. Pectins are readily soluble in water and they form colloidal suspensions. The suspensions set to form gels, a property that has been used in food industry to prepare food jellies.

c) Protopectins.

These are all the insoluble pectic substances. They are larger molecules than either pectic acid or pectins. They are also very much branched and found mainly in pears and apples.

iv) Pentosans.

These are polymers of five carbon sugars. The two commonly found pentosans are Xylans and Arabans.

a) Xylans.

Are the most commonly found pentosans. They are constituents of cell wall matrix and are formed from D-xylose units that are linked by β -1,4 linkages.

b) Arabans

These are relatively small polymers composed of L-Arabinose units. They are held by α -

1,5 linkages.

8.3.2 Lipids

a) Definition

Lipids is a group of organic compounds that have diverse structures. They have two common features:

They are present in living tissues and

They are soluble in organic solvents but insoluble in water.

Their solubility in organic solvents is a property that has been used in extraction of lipids from fresh and dried tissues. The organic solvents used are chloroform, acetone, ethers, certain alcohols and benzene. Good extraction can also be obtained from using mixtures. The mixtures commonly used are water/ethanol, water/acetone or chloroform/methanol (2:1v/v).

Lipids are fats or fat-like substances that are rich in carbon and hydrogen and poorer in oxygen. They are well distributed in the plant kingdom and are found in every cell. The lipid requirements of the cell are met within the cell itself because they are thought of as being immobile.

b) Classification

Lipids are divided into simple, compound and derived:

i) simple lipids

These are esters of fatty acids and alcohol. Lipids with glycerol as an alcohol are called fats and oils. Those with alcohol other than glycerol are called waxes.

ii) Compound lipids.

These are lipids that have fatty acids, alcohols and other groups. Examples are glycolipids, phospholipids and lipoproteins (have proteins).

iii) Derived lipids

They are lipids with high molecular weights. Examples are sterols and prostaglandins.

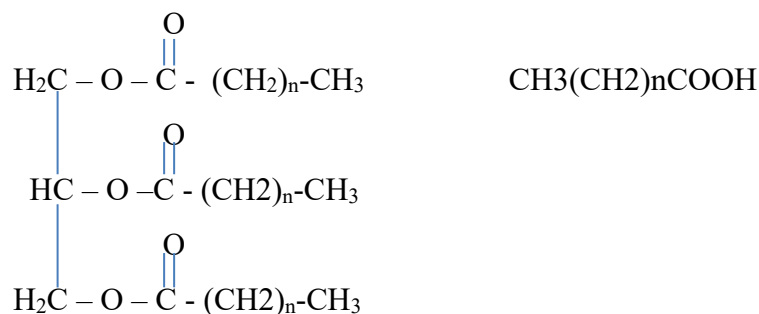
Fats and oils.

These two groups are chemically similar. The only difference is that fats are solids whereas oils are liquids at room temperature. Fats and oils are composed of long chain fatty acids whose carboxyl groups combine with hydroxyl groups of glycerol. As a result an ester is formed, which is an equivalent of a salt. Glycerol is a trihydric alcohol. If one hydroxyl group reacts with a fatty acid a monoglyceride is formed. Two hydroxyl groups react with two fatty acids to give a diglyceride. If all the three hydroxyl groups are used a

triglyceride is formed. Fatty acids can be saturated or unsaturated, that is, they have single and double bonds respectively.

Fats and oils are known as triglycerides.

The general structure of a triglyceride is:



Ester bond hydrocarbon chains.

The kinds of fatty acids determines the melting points and other properties of fats. The three fatty acids are different although two are occasionally identical. Oils have one to three double bonds in each fatty acid and this lowers the melting points and makes oils liquids at room temperature.

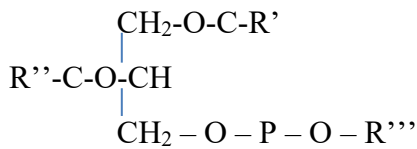
Waxes.

Waxes occurs in the cuticles of plants. They are complex mixtures of highly nonpolar lipids. The composition of this mixture is different in different plants. They usually contain esters, saturated long chain alcohols or aldehydes and saturated high molecular weight hydrocarbons. Esters predominate in the mixture.

Phospholipids.

These comprise a heterogenous group of compounds. They are important because they are associated with cell membranes. They are therefore present in every living cell as structural components of membranes. Phospholipids are widely distributed throughout the plant. As denoted by the name they contain phosphorus.

. They are diesters of phosphoric acid. The general structure is

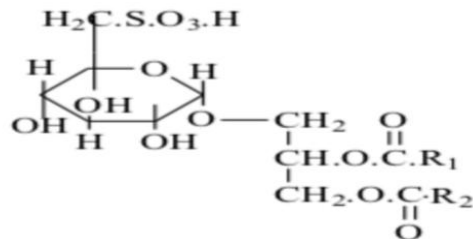


R', R'' and R''' are fatty acid residues.

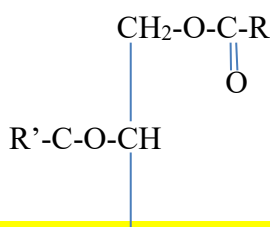
Glycolipids

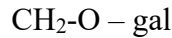
Glyco- relates to sugar. Glycolipids contain one or more monosaccharides incorporated in their structures. They are usually two types of these monosaccharides, glucose and galactose. Examples are sulpholipid (glucose) and monogalactosyl and digalactosyl glycolipids (one and two galactose residues respectively Sulpholipid (sulpho-6-deoxyglucose).

This glycolipid also contains sulphur.

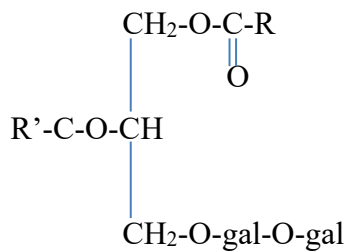


Monogalactosyl glycolipid. structure





Digalactosyl
glycolipid



Sterols

These are steroid alcohols. An example is cholesterol which is widespread in trace amounts in plants. Sistosterol, STIGMASTEROL.

Prostaglandins.

They have not been found in plants but it is only precursors that have been isolated in plants. They are mainly animal lipids that are basic to hormonal effects.

Fatty acids

Fatty acids are components of triglycerides, waxes, phospholipids and glycolipids. The carbon chain determines the chain length of the fatty acid, and hence the size of the fatty acid and most of the fatty acids have an even number of carbon atoms. Those having the shortest chain length) are caproic (C₆) caprylic (C₈) and capric (C₁₀). They are saturated

fatty acids and occur as minor components of the seeds of palmae.

Bonding between the carbon atoms is used to divide fatty acids into two: saturated and unsaturated fatty acids. Saturated fatty acids have only single bonds ($-\text{CH}_2-\text{CH}_2-$) and conform to the general formula, $\text{CH}_3(\text{CH}_2)_n \text{COOH}$. Unsaturated fatty acids have some double bonds ($-\text{CH}=\text{CH}-$) and do not conform to the general formula for saturated fatty acids. Examples of the fatty acids are given below.

* $\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$

Major saturated fatty acids

Name	N
Lauric	10
Myristic	12
Umaturic	14

Palmitic	14
Stearic	16
Arachidic	18
Behenic	20
Lignoceric	22

Palmitic acid is the most commonly occurring unsaturated fatty acids in plants.

Major unsaturated fatty acids

Name	Double bonds	Chain length
Palmitoleic	1	C ₁₆
Oleic	1	C ₁₈
Linoleic	2	C ₁₈
Linolenic	3	C ₁₈
Parinaric	4	C ₁₈

Oleic and linoleic acids are the most commonly occurring unsaturated fatty acids in plants. Oleic acid is abundant, widespread and mono-unsaturated(one double bond) fatty acid. Linolenic is the only important di-unsaturated fatty acid and is less widely distributed. Linolenic is the most common tri-unsaturated fatty acid.

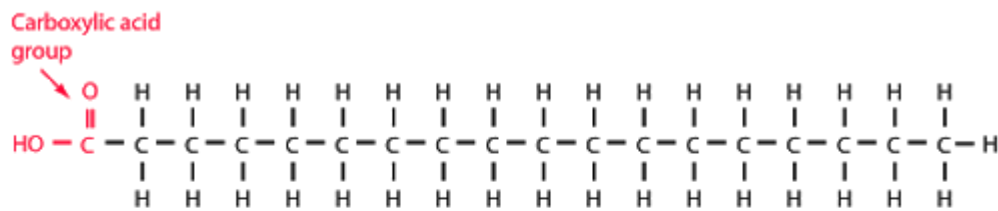
.

There are fatty acids that have unusual parts in their structures. They are the following:

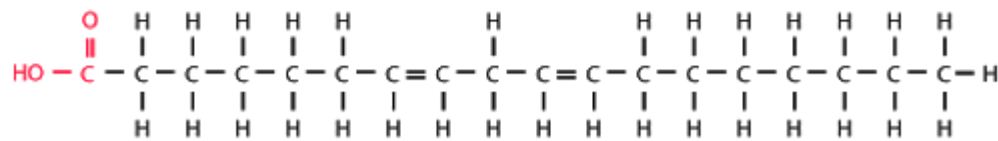
Name	Unusual part of structure	Description
Tariric	--C≡C---	Acetylenic double bond
Chaulmoogric	$ \begin{array}{c} \text{CH} = \text{CH} \\ \quad \quad \diagdown \\ \text{CH}_2 - \text{CH}_2 \quad \text{CH} \cdots \end{array} $	Ring structure

Ricinoleic	$\begin{array}{c} \text{---CH---} \\ \\ \text{OH} \end{array}$	Hydroxyl substituent
Vernolic	$\begin{array}{c} \text{---CH---CH---} \\ \quad \quad \\ \quad \quad \text{O} \end{array}$	Epoxy substituent
Licanic	$\begin{array}{c} \text{O} \\ \\ \text{---C---} \end{array}$	Keto substituent
Japanic	$\text{HOOC (CH}_2\text{)}_n \text{COOH}$	Additional carboxyl group

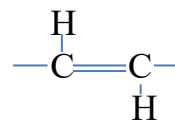
Lipid Structures



Stearic acid, an example of a saturated fatty acid

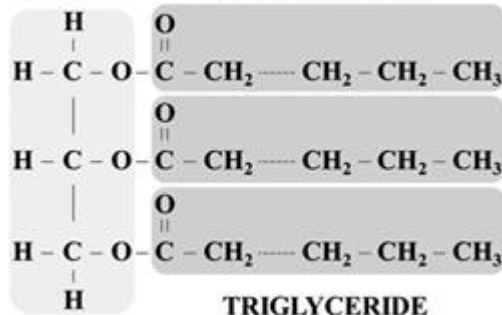


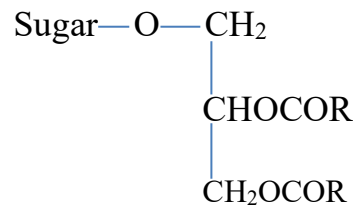
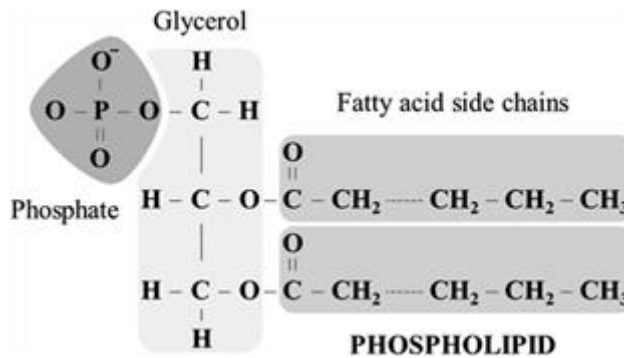
Linoleic acid, an example of an unsaturated fatty acid



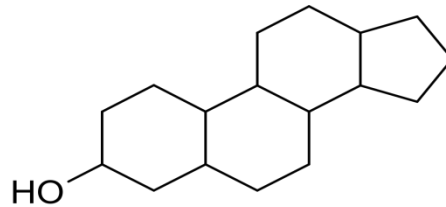
Glycerol

Fatty acid side chains





Glycolipid



Sterol

Amino acids

a) Definition and importance

Amino acids are small repeating units from which proteins molecules are formed. They are the building blocks of proteins. Acid hydrolysis of protein molecules yields amino acids. There are about 20 different amino acids found in proteins in plant cells.

Amino acids are important to plants in the following way:

- (i) Amino acids in plants varies from acting as osmolyte,
- (ii) They regulate ion transport,
- (iii) They modulate stomatal opening,

- (iv) They are involved in detoxification of heavy metals.
- (v) Amino acids also affect synthesis and activity of some enzymes,
- (vi) Amino acids act through a number of signaling pathways and mechanisms to mediate control of gene expression at the level of mRNA translation.
- (vii) Amino acids maintains redox-homeostasis that allows oxidation – reduction reactions to take place.
- (viii) Help to increase chlorophyll concentration in the plant leading to higher degree of photosynthesis.
- (ix) Protein biosynthesis - Amino Acids are fundamental ingredients in the process of Protein Synthesis. About 20 important Amino Acids are involved in the process
- (x) They play pivotal roles during signaling processes as well as in plant stress response.
- (xi) The degradation of amino acids can also contribute substantially to the energy state of plant cells under certain physiological conditions, e.g. carbon starvation.

a) Properties of Amino acids

1. Solubility

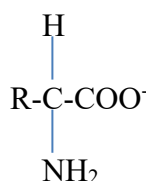
Amino acids are generally soluble in water and quite insoluble in non-polar organic solvents.

2. Melting points.

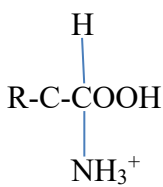
Amino acids have fairly high melting points and the melting is often followed by decomposition.

3. Amphoteric nature.

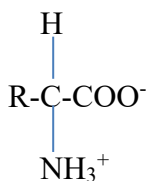
Amino acids possess both acidic (COOH) and basic (NH₂) groups. They are therefore capable of reacting with both alkalis and acids. Electrical charges on the amino acids influence their properties markedly. In solutions of varying pH (acidic-basic) an amino acid exists in different forms. In very basic solution the amino acid has NH₂ and COO⁻ functional groups and therefore acts as an anion. It has the following structure.



In very acidic solution (low pH) the amino acid has NH₃⁺ and COOH as the functional groups and so acts as a cation with the following structure



At a certain specific pH known as isoelectric point the amino acid has both positive (NH₃⁺) and negative (COO⁻) charges. This form is called zwitterion. It can function as an acid or a base and is said to be amphoteric in nature. Amino acids and proteins using this property are good buffers in living systems.

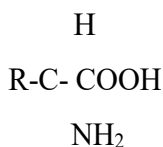


Since it has the two charges, it is dipolar and will exhibit net zero charge when placed in an electric field and it does not migrate. The migration property of amino acid cation and anion is used in a separation technique called electrophoresis.

b) Classification

Amino acids are generally divided into two groups: primary and secondary amino acids.

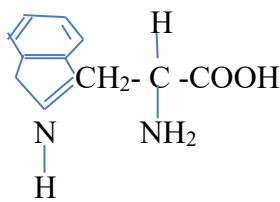
A primary amino acid has the following general structure.



Whereas NH_2 is the amino group, COOH the carboxyl group, H is hydrogen all of which are attached to the α carbon. Common properties of primary amino acids are attributed to the two groups and hydrogen. R is hydrogen, methyl group, straight or branched hydrocarbon chain with or without a benzene nucleus. Differences in the individual amino acid are due to R . Secondary amino acid does not conform to the general formula. Primary amino acids are further divided into aliphatic, aromatic, sulphur - containing acidic and basic. Examples of which are shown in the following table:

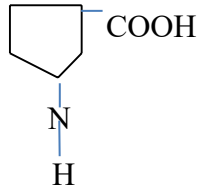
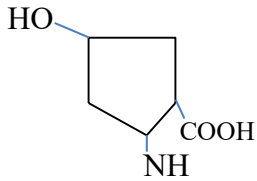
Primary Amino acids

Name	Formula	Types of amino acid
Glycine	$\begin{array}{c} \text{H} \\ \\ \text{H}-\text{C}-\text{COOH} \\ \\ \text{NH}_2 \end{array}$	Aliphatic
Alanine	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{C}-\text{COOH} \\ \\ \text{NH}_2 \end{array}$	Aliphatic
Valine	$\begin{array}{c} \text{CH}_3 \quad \text{H} \\ \diagdown \quad \\ \text{CH}-\text{C}-\text{COOH} \\ \diagup \quad \\ \text{CH}_3 \quad \text{NH}_2 \end{array}$	Aliphatic
Leucine	$\begin{array}{c} \text{CH}_3 \quad \text{H} \\ \diagdown \quad \\ \text{CH}-\text{CH}_2-\text{C}-\text{COOH} \\ \diagup \quad \\ \text{CH}_3 \quad \text{NH}_2 \end{array}$	Aliphatic
Isoleucine	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{CH}_2-\text{CH}-\text{C}-\text{COOH} \\ \quad \quad \\ \quad \text{CH}_3 \quad \text{NH}_2 \end{array}$	Aliphatic
Serine	$\begin{array}{c} \text{H} \\ \\ \text{CH}_2-\text{C}-\text{COOH} \\ \quad \\ \text{OH} \quad \text{NH}_2 \end{array}$	Aliphatic
Threonine	$\begin{array}{c} \text{H} \\ \\ \text{CH}_3-\text{CH}-\text{C}-\text{COOH} \\ \quad \\ \text{OH} \quad \text{NH}_2 \end{array}$	Aliphatic
Phenylalanine	$\begin{array}{c} \text{H} \\ \\ \text{C}_6\text{H}_5-\text{CH}_2-\text{C}-\text{COOH} \\ \\ \text{NH}_2 \end{array}$	Aromatic
Tyrosine	$\text{HO}-\text{C}_6\text{H}_4-\text{CH}_2-\text{C}-\text{COOH}$ $\begin{array}{c} \text{H} \\ \\ \text{C} \\ \\ \text{NH}_2 \end{array}$	aromatic

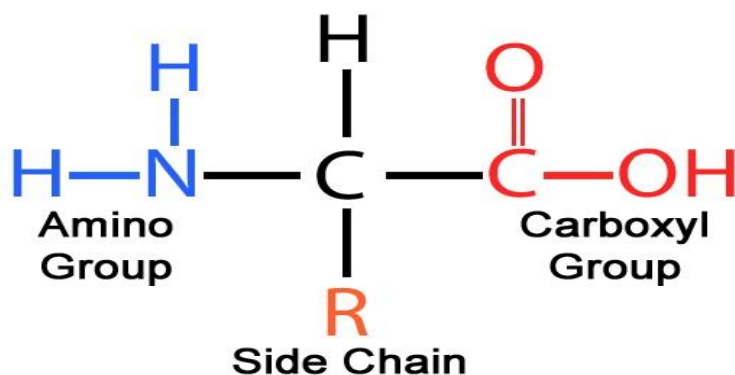
	NH ₂	
Tryptophan		Aromatic
Cysteine	$\text{HS}-\text{CH}_2-\overset{\text{H}}{\underset{\text{NH}_2}{\text{C}}}-\text{COOH}$	S-Containing
Methionine	$\text{CH}_3-\text{S}-\text{CH}_2-\text{CH}_2-\overset{\text{H}}{\underset{\text{NH}_2}{\text{C}}}-\text{COOH}$	S-containing
Aspartic acid	$\text{HOOC}-\text{CH}_2-\overset{\text{H}}{\underset{\text{NH}_2}{\text{C}}}-\text{COOH}$	Acidic
Glutamic acidic	$\text{HOOC}-\text{CH}_2-\text{CH}_2-\overset{\text{H}}{\underset{\text{NH}_2}{\text{C}}}-\text{COOH}$	Acidic
Histidine	$\text{N} \quad \text{NH} \quad \overset{\text{H}}{\underset{\text{NH}_2}{\text{CH}_2-\text{C}}}-\text{COOH}$	Basic
Arginine	$\text{H}_2\text{N}-\underset{\text{NH}}{\text{C}}-\text{NH}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\overset{\text{H}}{\underset{\text{NH}_2}{\text{C}}}-\text{COOH}$	Basic
Lysine	$\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\overset{\text{H}}{\underset{\text{NH}_2}{\text{C}}}-\text{COOH}$	Basic

Secondary amino acids

Name	Formula
------	---------

Proline	
Hydroxyproline	

General Structure of Primary Amino acid



8.3.3 Proteins

a) Definition and importance

Proteins are complex molecules that are formed by repeating units called amino acids.

These amino acids are linked together by peptide bonds connecting the carboxyl group of one amino acid with the amino group of the next amino acid.

The products of amino acids linked in this way are:

Dipeptide – consists of two amino acids. Tripeptide – consists of three amino acids.

Polypeptide - consists of a large number of amino acids.

About 20 different amino acids are present in a protein molecules. An amino acid may be present many times and occurs in different sequences. A protein molecule range in size from for example insulin (6000) to several millions.

Proteins are important to plants in the following ways:

- (i) Stored proteins carbon, nitrogen, and sulfur resources for subsequent growth and development. They are especially important in seeds, where they make up as much as 40 percent of the seed's weight and serve to store amino acids for the developing embryo.
- (ii) Certain proteins are enzymes that are used to trigger metabolic reactions within cells.
- (iii) Proteins form complexes that perform special functions in plant e.g. ferredoxin (iron-containing protein) plays a role in electron transport in photosynthesis
- (iv) Proteins are also important structural components of the cell wall.
- (v) Proteins serve as transport molecules

(b) Classification.

Proteins are classified, according to solubility properties and partly known chemical and physical properties, into simple and conjugated proteins.

i) Simple proteins:

When hydrolysed, simple proteins yield only amino acids. They have further been divided into six groups; albumins, globulins, glutelins, prolamines, histones and protamines. They are classified according to solubility in water, dilute salt solutions, ethanol, acidic, and basic solutions.

(a) Albumins:

They are soluble in water and dilute salt solutions. On exposure to heat they coagulate. An example is α -amylase of barley.

b) Globulins:

They are either insoluble or sparingly soluble in water but soluble in dilute salt solutions. They are coagulate when exposed to heat. Examples are storage proteins of seeds.

c) Glutelins:

They are insoluble in neutral solutions but soluble in weak acidic or basic solutions. They are chiefly found in cereal grains e.g. glutelin of wheat, oryzenin of rice.

d) Prolamines:

They are insoluble in water but soluble in 70%-80 % ethanol Prolamines are insoluble in absolute alcohol. When hydrolysed they yield relatively large quantities of proline and ammonia. Examples are zein of maize, gliadin of wheat and rice and hordein of barley.

e) Histones:

They are soluble in water and rich in basic amino acids. Histones are found in the cell nuclei where they are associated with nucleic acids.

f) Protamines:

They are soluble in water and rich in basic amino acids. They are also found in the nucleus where they are associated with nuclei acids. Tyrosine and tryptoplan are not found in these proteins and they have no sulfur.

(ii) Conjugated proteins:

These are proteins that have non-amino acid components known as prosthetic groups, in addition to amino acids or simple proteins. They are divided into nucleoproteins, glycoproteins, lipoproteins, chromoproteins and metalloproteins.

a) Nucleoproteins:

They yield a simple protein and a nucleic acid on hydrolysis.

b) Glycoproteins:

They contain small amounts of carbohydrates as prosthetic groups. They are found in the cell membranes.

c) Lipoproteins:

This group contains lipids as prosthetic groups. Example of lipids are

cephalin and lecithin. They are generally insoluble in water and are components of cell membranes.

d) Chromoproteins;

The prosthetic groups of these proteins are coloured compounds or pigments. Examples of chromoproteins are flavoproteins, carotenoid proteins, chlorophyll proteins and hemoglobins.

e) Metalloproteins.

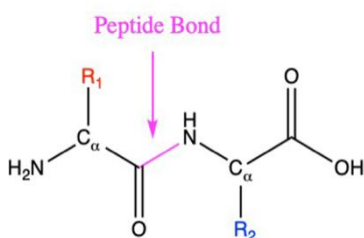
They contain metals as prosthetic groups. Many enzymes belong to this group and they require metals as activators.

c) Structural levels.

The biological properties of a protein molecule are related to the structure. There are four basic structural levels in a protein molecule: primary, secondary, tertiary and quaternary.

i) Primary structure:

This is obtained through the formation of a definite linear sequence of amino acids, held together by peptide bonds. What is formed is a polypeptide. Many proteins contain more than one polypeptide chains, Another characteristic of the primary structure is the presence of a disulfide (-S-S-) bond. This bond is established between two S- containing amino acids that are close together be they of the same polypeptide or two separate polypeptides.



ii) Secondary structure:

This is a spirally coiled structure of the primary structure or polypeptide chain. The spiral or helical arrangements are maintained by hydrogen bonds. Salt links and Van der Waals forces are also found in the secondary structure. The structure is associated with biological function of the protein molecule. The breakage of these bonds by mild conditions dissociates the protein molecule.

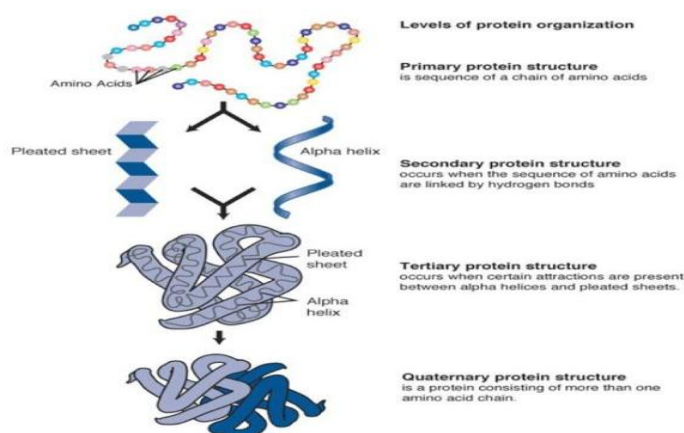
iii) Tertiary structure:

This structure is formed by the folding and superfolding of secondary structure. It is maintained by hydrogen bonding, salt links and Van der Waals forces. There is also an interaction of R-groups. The tertiary structure is also associated with biological function of a protein molecule eg. Enzyme activity. Factors such as extreme temperatures, pH, ultra violet radiation cause disruption of this structure.

iv) Quaternary structure

This is formed from the folding of the tertiary structure giving the three dimensional shape of a protein molecule. All bonds in the previous structures are found in this

structure. In addition there is an association of hydrophobic groups that combine excluding water.



8.4 Summary

Carbohydrates are polyhydroxyaldehydes, polyhydroxyketones and their derivatives. They provide energy, used in structure and provide skeletons for synthesis of other organic compounds. They are divided into monosaccharides, oligosaccharides and polysaccharides.

Lipids are fat or fat-like substances. They have been classified into simple lipids which consists of fats, oils and waxes; compound lipids which include glycolipids, phospholipids and lipoproteins. Derived lipids is a third class that include sterols and prostaglandins. Fatty acids are the building blocks of both simple and compound lipids. There are two types of fatty acids. saturated (single bonds) and unsaturated (double bonds).

Amino acids are building blocks of proteins. They are divided into primary and secondary amino acids. Primary amino acids have further been divided into aliphatic, aromatic, sulphur-containing, acidic and basic amino acids. They have both basic and acid groups, giving the amphoteric nature.

Proteins are complex compounds that have primary, secondary, tertiary and quaternary structural levels. They have been broadly divided into simple and conjugated proteins. These have been further grouped according to solubility properties for simple and prosthetic groups for conjugated proteins.

8.5 Activity

Carbohydrates

- a) Put 20 gm of chopped carrot into a boiling tube. Cover with water and boil. Transfer the liquid to a second boiling tube and add equal quantities of Fehlings reagent. Observe a red precipitate is formed by reducing sugars.
- b) Cut soaked maize grains vertically. Apply a little iodine potassium iodine solution to cut surface. Note the colour. Starch forms blue black

8.6 Questions;

1. Define the following: (a) carbohydrates (b) lipids (c) amino acids (d) proteins.
2. Outline the significance of carbohydrates to plants.
3. Write short notes on structures found in carbohydrates.
4. Briefly describe pectic substances.
5. Distinguish between: (a) simple and compound lipids (b) phospholipids and glycolipids. (c) saturated and unsaturated fatty acids.
6. How are triglycerides formed?
7. Identify the functional groups of an amino acid.
8. Discuss the classification of primary amino acids.
9. Explain why amino acids and proteins are good buffers.
10. Explain how a peptide bond is formed.
11. Discuss the classification of proteins.

8.5. Activity**Properties of amino acids****a) solubility**

Note the appearance of glycine and tyrosine amino acids. Test their solubilities in water. Determine solubility in 2 M HCL and 2M NaOH.

b) Determination of pH and Isoelectric point precipitation.

Determine the pH of the solution or suspension obtained in (a) above. For isoelectric point dissolve about 0.5g of tyrosine in 1 ml of water containing 2 drops of 2 m NaOH (Warm if necessary) and add 1 drop of methyl red indicator. Add 0.2 M HCL dropwise with constant shaking until the indicator changes colour. Repeat the experiment by dissolving 0.5 g of tyrosine in 1 ml of water containing 2 drops of 2 m HC. Add a drop of methyl red indicator. Add 0.2 M NaOH dropwise shaking constantly until the indicator changes colour. Record the pH.

LESSON 9: ENZYMES

9.1 Introduction

In the previous chapter we learnt about proteins. Enzymes are partly or entirely proteins. Long before enzymes were isolated and identified as proteins, human beings had been using them. Their importance can be traced to the ancient Greeks who use them in the process of fermentation in the production of wine. Other practices that require the use of enzymes were making of cheese vinegar and the leavening of bread.

More information was subsequently obtained due to the need to improve the quality of these products. Juice of ground and pressed yeast was found to ferment sugar. Intact living cells were reported to be having this ability of enzymatic action. Following the isolation of urease enzyme, it was reported that enzymes are proteins.

In this chapter we will consider the nature nomenclature and classification, catalytic action distribution and the factors that affect enzyme activity.

9.2 Objectives:

By the end of this topic on enzymes students should be able to:

- a) define the word enzyme
- b) name and classify the numerous enzymes found in plant cells.
- c) associate their activity and distribution.
- d) identify their various properties.
- e) explain how the enzymes work.
- f) understand the factors that regulate enzyme activity

9.3 Content

9.3.1 Definition

Substances that increase the rate of chemical change without themselves appearing to change in the reactions are called catalysts. There are both inorganic and organic catalysts.

The word enzyme was coined by W. Kuhne in 1878. It is a Greek word derived from: “En” which means “in” and “Zyme” which means “leaven”. It therefore was coined from the work of fermentation and leavening of bread.

Enzymes are defined as simple or compound proteins that act as specific organic catalysts. Many biochemical reactions in living systems are under the regulation of enzymes. They are active in practically all cellular processes. Enzymes help to control the precision and the rate of changes taking place in the protoplasm. They are involved in the processes of building and degradation.

9.3.2 Nomenclature and classification.

Enzymes are named by adding -ase at the end of a word.

They are named generally according to the substrate they catalyze or the type of reaction they participate in. Substrates can be individual substances or groups of substances.

Examples are

i) individual substances:

arginine – arginase

Tyrosine – tyrosinase.

ii) group of compounds

lipids – lipases

proteins – proteinases

carbohydrates – carbohydrases.

As far as the type of reaction catalyzed is concerned, the nomenclature is exemplified by the following:

Carboxylation – carboxylases

Phosphorylation – phosphorylases

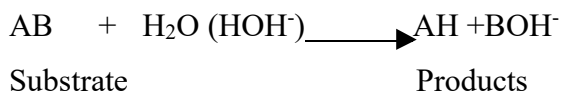
Hydrolysis – hydrolases

Dehydrogenation – dehydrogenases.

The classification system given by the Commission on Enzymology in 1961 under the direction of the International Union of Biochemistry has six main divisions of enzymes. They are classified according to the biochemical reaction catalyzed. These divisions are:

1. Hydrolases

They catalyze the addition of water across a specific bond of the substrate before the cleavage of the substrate before the cleavage of the substrate. The general reaction catalyzed is:



Examples are carbohydrases, esterases, proteases and phosphatases.

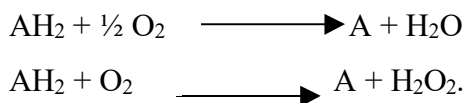
2. Oxidoreductases.

These are enzymes that catalyze the removal or addition of hydrogen, oxygen or electrons from a substrate which in turn becomes oxidized or reduced.

Examples are:

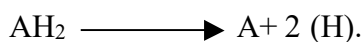
i) Oxidases

Oxalate oxidase belong to this group. They transfer hydrogen from the substrate to molecular oxygen to form either water or hydrogen peroxide.



ii) Dehydrogenases

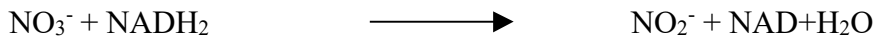
Examples are succinic dehydrogenase, lactic dehydrogenase, alcohol dehydrogenase and malic dehydrogenase. They remove hydrogen from the substrate.



iii) Reductases

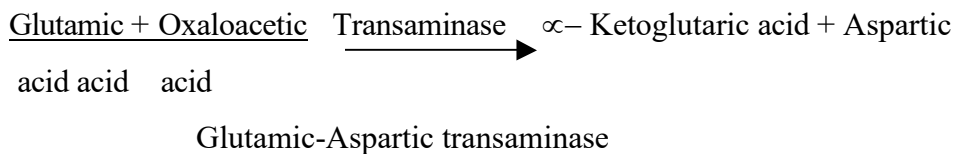
They are responsible for addition of hydrogen or electron and removes oxygen.

Nitrate reductase is an example.



3. Transferases.

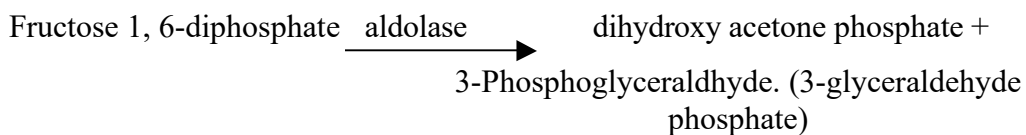
They transfer a group from a donor molecule to an acceptor molecule. Examples are transaminases and kinases. A specific reaction catalysed by a transferase is:



In this reaction an amino group is transferred from glutamic acid to oxaloacetic acid which is converted to aspartic acid.

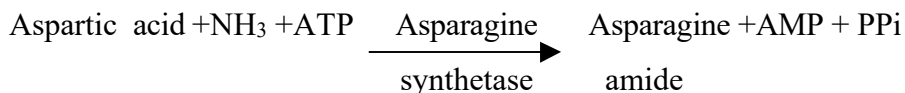
4. Lysases.

These enzymes remove groups from a substrate without hydrolysis examples are fumarases, carboxylases and aldolases. A specific reaction that exemplifies is:



5. Ligases or synthetases

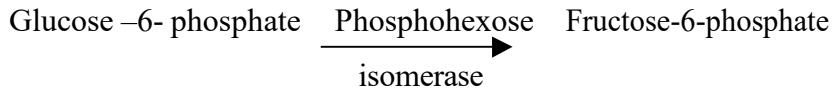
The enzymes catalyses the synthesis of various types of bonds e.g. C-C, C-N, C-S and C-O. Examples are thiokinase and synthetase.



6. Isomerases.

These are enzymes that catalyse isomeric changes (isomerization). Isomers are having identical molecular formulas (same number of atoms of each element) but distinct

arrangements of atoms in space. Examples of enzymes are isomerases, epimerases and racemases. A specific reaction that exemplifies the action of a member of this group is:



9.3.3 Distribution of enzymes

Enzymes are compartmentalized within the plant. This enhances the association of the enzyme and the substrate. They are restricted to the area where they are in intimate contact with the substrates.

Enzymes are divided in two according to their compartmentalization. These intracellular and extracellular enzymes.

Intracellular are formed within the cell walls of the cells. They are found in the cytoplasm and in organelles. These are:

a) Cytoplasm:

The enzymes found in the cytoplasm are numerous. These include enzymes of glycolysis and Hexose monophosphate shunt. Various hydrolytic enzymes and phosphorylases are also found in the cytoplasm.

b) Mitochondria:

Enzymes necessary for the complete oxidation of pyruvate (pyruvic acid) to CO_2 and H_2O are found in the mitochondria. Along with these are enzymes necessary for the electron transfer to O_2 via electron transport system that results in ATP formation.

c) Chloroplast:

Enzymes necessary for the dark reactions of photosynthesis (CO_2 assimilation) are present in the matrix of the chloroplast. Cytochrome enzymes are also located in the chloroplast and their activity leads to the production of ATP. The enzymes involved in the synthesis of photosynthetic pigments (chlorophylls and carotenoids) are also found in the chloroplast.

d) Nucleus:

Not many enzymes are found in the nucleus. Deoxyribonuclease is located in the nucleus.

This enzyme catalyzes the cleavage of DNA.

Extracellular enzymes are those located in the environment external to the cell. Examples include enzymes involved in extracellular digestion and transport of nutrients into the cell.

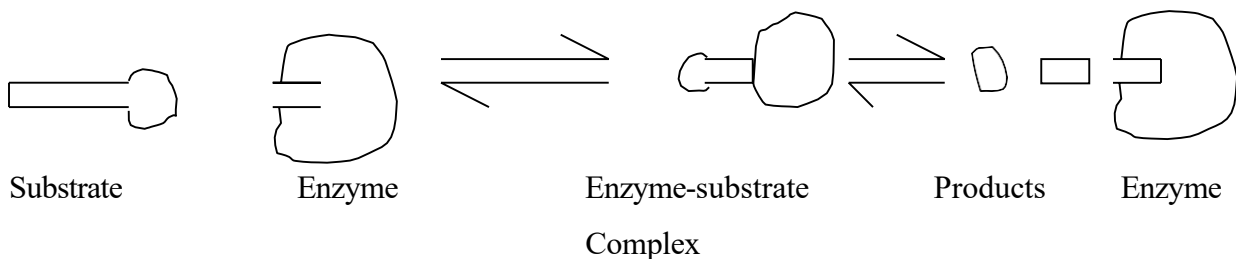
9.3.4 Properties of Enzymes.

1. Enzymes are colloidal in nature. They therefore provide a large surface area for reactions to take place.
2. Enzymes can react with both acidic and alkaline substances (cations and anions).
3. Enzymes show sensitivity. They can be precipitated and denatured by factors that inactivate proteins.
4. Enzymes are thermolabile. They are heat sensitive and get denatured thus become inactive at extremely high temperatures.
5. Enzymes can be inhibited. They have active sites which can be occupied by other substances referred to as competitors at the expense of the substrates. Some substances affect the structure of enzymes that lead to reduced activity.
6. Enzymes have activators which accelerate enzyme activity. These activators can function as protectors or anti-inhibitors.
7. Enzymes hasten the completion of a reaction but will not affect the equilibrium of that reaction. They speed up the attainment of the equilibrium of a reversible reaction which would otherwise be slow.
8. Enzymes are greater in size than the substrate molecule. They therefore have active sites that take up more substrate molecules that are smaller in size.
9. Enzymes are specific. Their catalytic action is specific. Numerous enzymes have been isolated and found to be specific. They react with only particular substrate. This property is very strong for some enzymes but general in others. There are two types of specificity:
 - i) Group specificity (absolute specificity) shown by enzymes that affect a general group e.g. kinase which phosphorylates hexoses.
 - ii) Stereospecificity: shown by enzymes that affect D or L optical or structural isomers.

10. At the end of the reaction, enzymes remain unaffected by the reaction catalyzed. They are close to true catalysts. True catalysts remain unaffected at the end of a reaction. The activities of enzymes are confined to narrow ranges of temperature, pH etc.
11. Enzymes are active in small amounts. A small amount of enzyme converts a large amount of substrate to product. A substrate is the starting material in a biochemical reaction. A turnover number is the number of moles of substrate converted per minute by a mole of enzyme. This can be calculated for each reaction catalyzed.

9.3.5 Catalytic action

Fisher (1898) proposed a “key-lock theory” to explain the mode of action of enzymes. Just like a particular lock can be opened by a particular key, a particular enzymes acts of a particular substrate. Another theory put forward by Michaelis and Menten (1913) attributed the catalytic action to the fact that enzymes have certain active sites for the attachment of substrate. An enzyme forms a weakly bound compound with substrate which on hydrolysis decomposes into the reaction products.

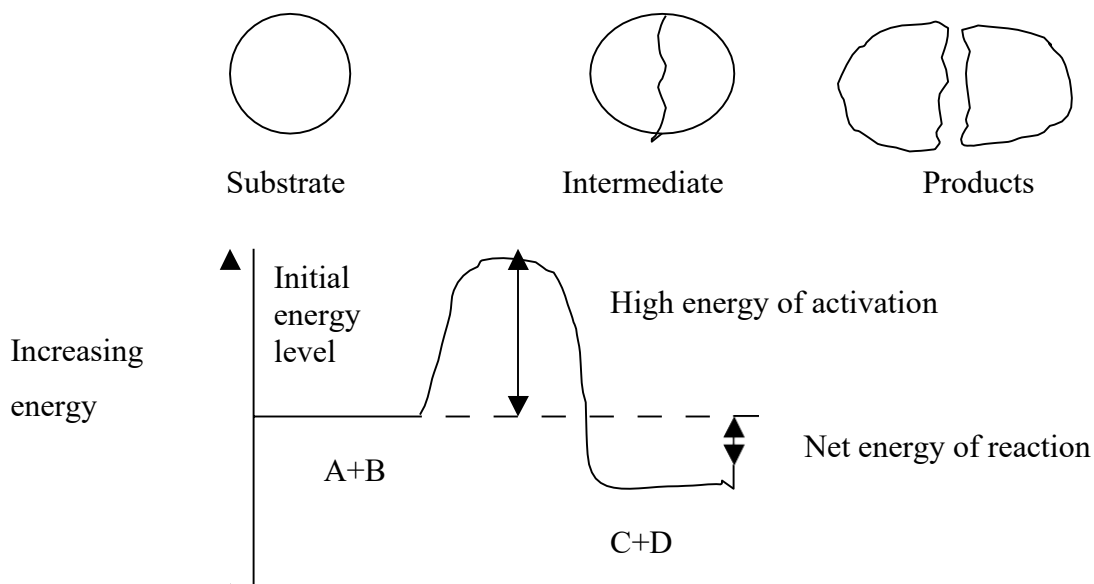


The theory depends upon physical contact between substrate and molecules of the enzyme. It is supported by competitive inhibition. The active site of the enzyme is blocked by another substance which has nearly about the same structure as that of substrate. The activity of the enzyme is lost as a result.

9.3.6 Activation Energy

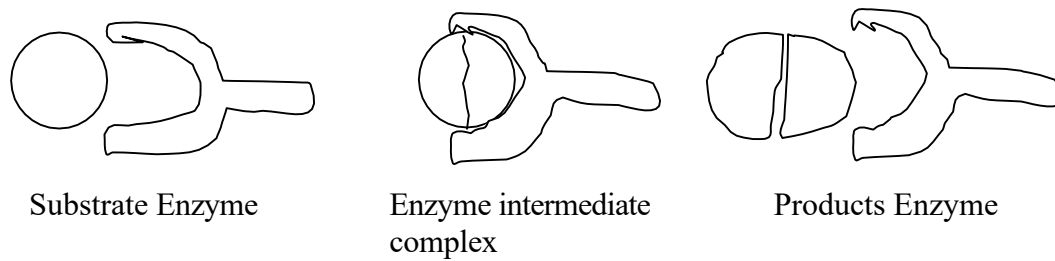
At a specific temperature, the enzymes affect the rate of a reaction by lowering the activation energy. In a reaction there are reactants (substrates) and products. These can be represented A and B for reactants and C and D for products. A and B have certain kinetic energy and before they are converted to C and D they must surmount an energy barrier. The kinetic energy of A and B is higher than that of C and D. The energy required to overcome the energy barrier is called activation energy of the reaction.

This can be illustrated using a catalysed and uncatalysed reaction. In uncatalysed reaction (ordinary reaction) the rate is slow and the rate of formation of an intermediate is limited and a high energy of activation is required.

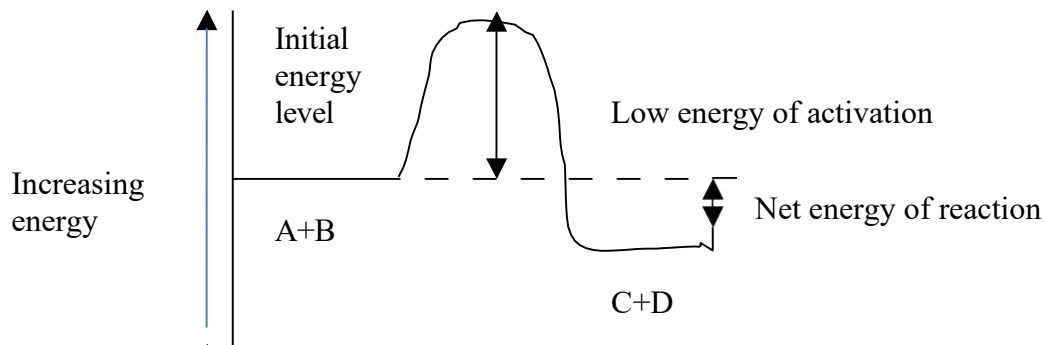


Uncatalysed reaction.

Prosthetic groups, c



mmmm



Catalysed reaction.

9.3.7 Prosthetic Groups, Cofactors and Coenzymes.

Enzymes are proteins. Proteins are divided into two: simple and conjugated proteins: A conjugated protein has a part that consists of only amino acids. This is called apoenzyme. The other part is non-protein and is called prosthetic group. The prosthetic group can be inorganic or organic.

Inorganic prosthetic groups are called cofactors. They can be metal ions e.g. copper, iron, magnesium, manganese, zinc, calcium, potassium and cobalt. Non-metal ions include chloride ions. They are also called activators. If they are removed from the enzyme it loses its catalytic activity which is restored when returned to the apoenzyme.

Organic substances associated with enzymes are called coenzymes. They can associate or dissociate readily from the enzyme. Removal of coenzymes reduces the catalytic activity of the enzymes. Their role is to act as donors or acceptors of groups of atoms or electrons added to or removed from the substrate. Examples are Nicotinamide adenine dinucleotide

(NAD), Nicotinamide adenine dinucleotide phosphate (NADP), Coenzyme A (CoA), Flavin mononucleotide (FMN), and Flavin adenine dinucleotide (FAD). Some coenzymes contain vitamins which are organic compound synthesized in plants that function in minute quantities. Pyridoxal phosphate, Coenzyme Q (CoQ)

9.3.8 Factors affecting enzyme activity.

1. Substrate concentration:

The enzyme molecule is larger than the substrate molecule and has several active sites on the surface. When the substrate concentration is low some of the active sites are not occupied because the substrate molecules are scattered and collisions with enzyme are few. The enzyme is not working at maximum efficiency.

When the substrate concentration is increased enzyme activity also increases because the number of substrate molecules in the immediate vicinity of the enzyme active sites are increased.

The chances of them coming in contact with the active sites increases. At constant enzyme concentration, an increase in substrate concentration will increase the rate of enzyme - catalysed reaction.

When all the active sites have been occupied then the enzyme works at maximum efficiency.

2. Enzyme concentration:

If all the active sites on the enzymes have been utilized after increasing the substrate molecules, additional substrate will not increase the rate of reaction. When the enzyme concentration is increased the rate of reaction increases because more active sites are made available to the excess substrate molecules.

3. Temperature:

Enzyme catalysed reaction just like any chemical reaction is affected by temperature. At 0°C the rate of reaction is practically zero. The rate increases upto about 25°C. Conditions of denaturation of enzymes begin to appear at 30°C. At between 35°C and 60°C the enzymes loses their catalytic properties because of denaturation. Temperature affects enzyme catalysed reaction in two ways by:

- i) increasing the kinetic energy of both substrate and enzyme molecules.
- ii) Increasing chances of contact.

4. pH

pH affects enzyme activity by:

- a. denaturing the enzyme molecule causing a fall in its activity.
- b. affecting ionic groups which are present on enzymes since they are proteins. These groups may be charged or uncharged depending on the pH of the environment. The pH affect enzyme activity particularly if these ionic groups are part of the active sites.

Enzymes have their optimal pH.

5. Inhibitors.

These are chemical substances that reduce enzyme activity. There are two types: competitive and non competitive inhibitors.

- i) Competitive inhibitors.

These are structural analogues of the substrates. They compete for the same active sites occupied by substrates. This can be overcome by adding more substrate molecules, to increase chances of contact. Therefore competitive inhibition can be reversed.

- ii) Noncompetitive inhibitors.

These reacts with either parts of the enzyme or enzyme substrate complex resulting in the alteration of the structure that is essential for catalytic activity. This type of inhibition cannot be solved by adding more substrate molecules.

9.4 Activity

Extraction and testing of amylase enzyme.
Place about 50 germinating wheat grains in a mortar with a little clean moist sand. Grind thoroughly and add four volumes of distilled water. Allow to sand for two hours then filter. The filtrate contains the enzyme in crude form. Put 5 ml of the filtrate in each of two test tubes. One is heated to boiling temperature. 5 ml of freshly prepared starch solution is added to each test tube containing the crude enzyme. Place a few drops of the mixture from each test tube on a spot plate and test for starch with iodine solution (I_2KI).

9.5. Summary

Enzymes are conjugated proteins with cofactors and coenzymes which are known as prosthetic groups. There are numerous enzymes that catalyse many reactions in plant cells. They have been named by adding -ase at the end, and based on substrates and reactions catalysed. The various divisions of enzymes are hydrolases, (oxidases, dehydrogenases and reductases) transferases, lysases, ligases and isomerases.

Enzymes are found within and outside the cell. They are colloidal, sensitive, thermolabile, inhibited, specific and are active in small amounts. They also react with acids and bases, hasten completion of reactions, greater in size than substrate molecules and remain unaffected at the end of the reaction.

9.6 Questions.

1. State the criteria used to name enzymes and show how they have been used in the nomenclature of enzymes.
2. Write short notes on oxido reductases.
3. Distinguish between: (a) lysases and ligases. (b) intracellular and extracellular enzymes.
4. Give an account of the distribution of enzymes in a plant tissue.
5. Discuss the properties of enzymes.

LESSON 10: PLANT GROWTH SUBSTANCES

10.1 Introduction

In this lesson you will learn the five groups of plant hormones and their effects on plant growth and development.

10.2. Objectives

By the end of the lesson you should be able to:

specify the different classes of endogenous plant hormones.

describe their effects on plant growth.

10.3. Content.

Definition: A plant hormone is an organic compound synthesized in one part of the plant and translocated to another part where, in very small concentrations it causes a physiological response.

Characteristics

Plant growth hormones, plant growth substances or plant growth regulators are of two types : plant growth promoters (auxins, gibberellins and cytokinins) and plant growth inhibitors (abscisic acid and ethylene). These substances exhibit the following characteristics:

Differentiation and elongation of cells.

Formation of leaves, flowers, and stems.

Wilting of leaves.

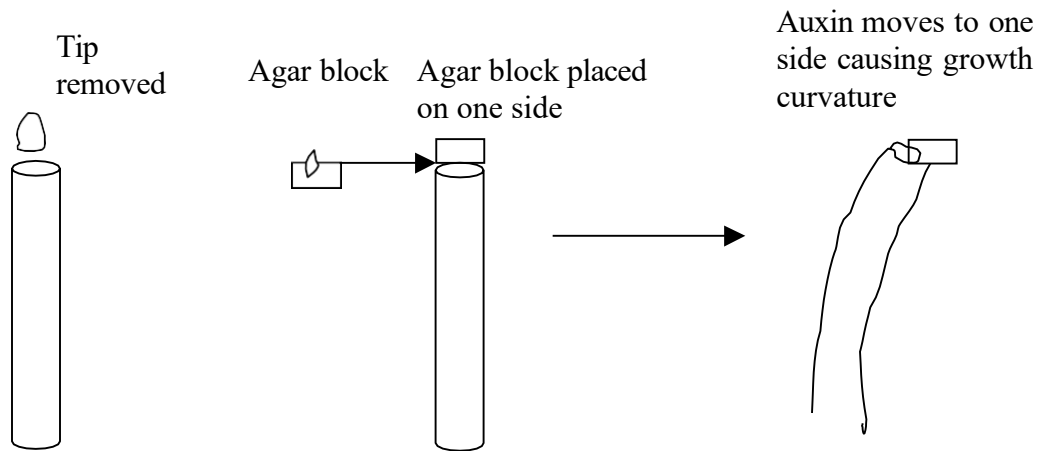
Ripening of fruit.

Seed dormancy, etc.

10.3.1 The Auxins.

- Frits Went discovered in 1926 that some compound caused the curvature of oat

- (*Avena*) coleoptiles toward light (phototropism) and called the compound auxin.
- He demonstrated that auxin present in the oat coleoptile tips could diffuse into agar blocks. The activity of this auxin was detected by curvature of the coleoptile caused by enhanced elongation on the side to which the agar block was applied.

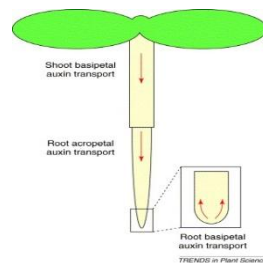


Went's demonstration of auxin in the Avena coleoptile.

Went's auxin is called indole acetic acid (IAA). Other naturally occurring auxins are phenylacetic acid (PAA) and indole-3-butyric acid (IBA)]. Similar compounds synthesized by chemists cause many of the physiological responses common to IAA e.g naphthalene acetic acid (NAA), 2,4- dichlorophenoxyacetic acid (2,4-D) and 2,4,5 – trichlorophenoxyacetic acid (2,4,5-T), Indole-3-propionic acid and Indole-3-pyruvic acid. These are considered to be auxins but because they are not synthesized by plants they are not hormones. They are referred to as plant growth regulators.

Transport of auxin.

Auxins are synthesized in stem tips and young leaves. Auxin transport is always polar i.e. they move down the stem toward the base i.e basipetal direction. Movement in the root is also polar but acropetal i.e. toward the apex. Auxin transport requires energy.



Effects of auxins

i) Apical dominance.

In stems of most species, the apical bud exerts an inhibitory influence i.e apical dominance on the lateral buds preventing or slowing their development.

When the apical bud is removed the lateral buds sprout into branches. If an auxin is added to the stump after removal of the apex, the lateral bud development is inhibited. This indicates that the inhibitory compound produced by the apical bud is an auxin.

(ii) Root initiation.

In many species removal of buds and young leaves which are rich in auxins inhibits the formation of lateral roots. Substitution of these organs by auxins restores the root-

forming capacity. This is evidence that auxins from stems influence root initiation.

(iii) Root elongation

Auxins promote elongation of sections excised from roots but at very low concentrations. Higher concentrations inhibit root elongation.

(iv) Flowering - Auxin has long been implicated in flower development. It is necessary for the initiation of floral primordia thereby facilitating flowering in plants. This has been supported by application of auxin to pineapple promoting flowering.

(v) Propagation – Auxins are involved in the initiation of roots in stem cuttings therefore used in the process of plant propagation.

(vi) Herbicides - Auxins are widely used as herbicides to kill dicot weeds therefore used by gardeners to keep lawns free from weeds. Auxin herbicide action leads to synthesis of ethylene and ABA. ABA directly inhibits cell division and expansion and promotes, together with ethylene, foliar senescence with chloroplast damage and destruction of membrane and vascular system integrity. Growth inhibition, tissue desiccation and decay and finally plant death are the consequences.

Prevention of dropping of leaves and fruits at early stages.

Regulate xylem differentiation and assists in cell division.

Used to produce fruit without preceding fertilization.

Promote natural detachment (abscission) of older leaves and fruits.

10.3.4 The Gibberellins/Gibberellic acids.

Discovered in Japan in studies with diseased rice plants that grew so tall that they could not support themselves (lodge). The Japanese called the disease bakanae (foolish seedling) disease. It was caused by the fungus *Gibberella fujikoroi*.

- extracts from the fungus applied to rice plants caused the same symptoms as the fungus itself. This showed that a chemical substance was responsible for the disease. The active compound from the fungus was isolated and named gibberellin.
- More than 60 gibberellins have been discovered. They are abbreviated GA with a subscript e.g. GA₁ GA₂, GA₃ to distinguish them.

Effects of Gibberellins

- i) Promotion of growth of intact plants.
 - Gibberellins enhance elongation of intact stems more than they do excised stem sections. Most dicots and some monocots grow faster when treated with gibberellins. Species in the rosette form that have short internodes e.g. cabbage grow 2 m tall and flower after GA application. The untreated plants remain short and vegetative.
- ii) Promotion of germination of dormant seeds and growth of dormant buds.
 - Seed and bud dormancy is broken by winter cold periods
 - Bud dormancy can be also broken by increasing day lengths.
 - In seeds of many species dormancy is broken by brief periods of red light when the seeds are moist.
 - Gibberellins overcome both seed and bud dormancy acting as a substitute from low temperature, long days and red light.
- iii) Flowering

- Some plant species flower only if the light period (Photoperiod) exceeds a critical length. Others flower if the photoperiod is shorter than a critical length.

Gibberellins can substitute for the long-day requirement. Gibberellins also overcome the need some species have for an inductive cold period to flower (vernalization).

iv) Parthenocarpy

In some plant species gibberellins sprayed on unpollinated flowers results in seedless (parthenocarpic) fruit development.

Functions of Gibberellins

Delay senescence in fruits.

Involved in leaf expansion.

Break bud and seed dormancy.

Promote bolting in cabbages and beet.

Facilitate elongation of fruits such as apples and enhance their shape.

Used by the brewing industry to accelerate the malting process.

Used as the spraying agent to increase the yield of sugarcane by elongation of the stem.

In young conifers, utilized to fasten the maturity period and facilitate early seed production

Helps in increasing the crop yield by increasing the height in plants such as sugarcane and increase the axis length in plants such as grape stalks.

Gibberellins are acidic in nature.

It also delays senescence.

10.3.5. The cytokinins.

In 1913 G Haberlandt discovered that a compound present in phloem of various plants stimulated cell division. Compounds that stimulate cell division (cytokinesis) are known as cytokinins. Zeatin is a cytokinin found in the milky endosperm of maize (*Zea mays*).

Cytokinins are synthesized in young organs e.g. seeds, fruits, leaves and in root tips.

Effects of cytokinins.

i) Delay of senescence

When a mature leaf is detached, it ages i.e it senesces more rapidly than it would if it were attached.

In dicot leaves adventitious roots form at the base of the petiole and senescence of the leaf is delayed.

Cytokinins replace the need for roots in delaying senescence (ageing) through promoting nutrient mobilization. This indicates that the roots provide cytokinins that delay leaf senescence. This has can be demonstrated by fresh leaf crops like cabbage and lettuce. They are used to keep flowers fresh for a longer time.

ii) Increase of nutrient sink activities.

Cytokinins cause transport of many solutes from older parts of the leaf and from older leaves into the zone where a cytokinin has been applied i.e. treated zone.

If one part of a leaf is treated, radioactive metabolites added to an adjacent leaf migrate via phloem into the treated zone and accumulate there. Because young leaves

are rich in cytokinins, they remove nutrients from older leaves. In this way cytokinins enable young tissue to act as sinks for metabolites.

iii) Promotion of lateral bud development of dicots.

If a cytokinin is applied to a dormant lateral bud dominated by the apical bud above it, the lateral bud begins to grow. Functions of Cytokinins

Break bud and seed dormancy.

Promotes cell division and can be used in tissue culture to induce cell division in mature tissues.

Facilitate adventitious shoot formation and lateral shoot growth.

Involved in the formation of new leaves and chloroplast organelles within the plant cell.

Used to induce the development of shoot and roots along with auxin, depending on the ratio.

10.3.6. Ethylene.

Ethylene is a gaseous hormone that is produced in leaves, fruits and roots. It causes leaf senescence and abscission.

Effects of ethylene.

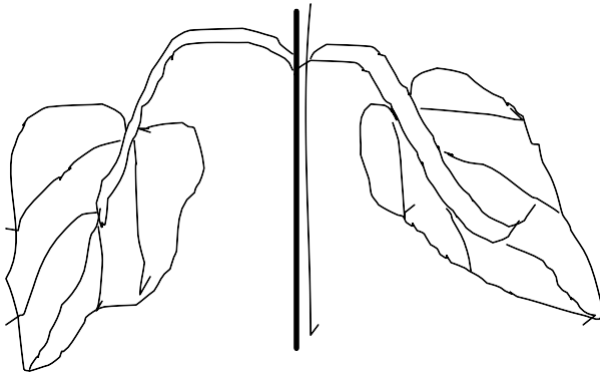
(ii) Plants in waterlogged soils.

Waterlogged soils rapidly become hypoxic (oxygen deficient) because water fills the air spaces. Oxygen supply to the roots is therefore cut off. However, ethylene that is present is trapped in the roots because its escape through the water is reduced. As a result

cellulase, an enzyme that breaks down cell walls is synthesized in the cells of the cortex.

The cortex cells disappear resulting in air-filled aerenchyma tissue.

In the well-aerated shoots ethylene is synthesized. This ethylene causes leaf epinasty (leaf bends downward) by promoting elongation of cells on the upper side of the petiole while the cells on the lower side do not elongate.



Leaf epinasty

ii) Elongation of stems and roots.

- Ethylene inhibits elongation of stems and roots in dioct. When elongation is inhibited the roots and stems become thicker by enhanced radial expansion of the cells.

iii) Fruit ripening.

- Ethylene stimulates fruit ripening. In stored fruits the ethylene produced by each fruit has a cumulative effect and stimulates other fruits to ripen faster and to produce more ethylene. Thus ripening become autocatalytic. (NB: autocatalysis is the catalysis of a reaction by a product of that reaction).

Functions of Ethylene

Ethylene is the most widely used plant growth regulator as it helps in regulating many physiological processes.

Induce flowering in the mango tree.

Promotes sprouting of potato tubers.

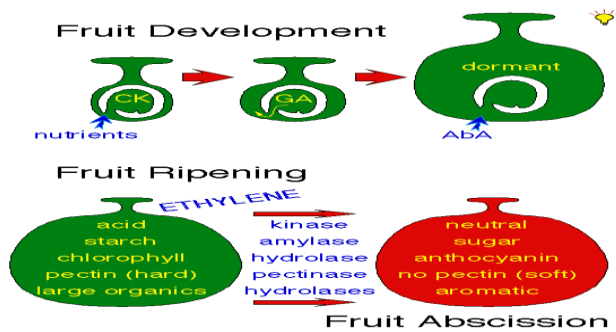
Breaks the dormancy of seeds and buds.

Enhances respiration rate during ripening of fruits.

Applied to rubber trees to stimulate the flow of latex.

Facilitates senescence and abscission of both flowers and leaves.

Used to stimulate the ripening of fruits. For example, tomatoes and citrus fruits.



Affects horizontal growth of seedlings and swelling of the axis in dicot seedlings.

Increases root hair formation and growth, thus aids plant to expand their surface area for absorption.

10.3.7. Absciscic Acid (ABA)

- ABA is the most widespread inhibitor in plants.
- It was independently discovered by P F Wareing and co-workers in the UK who called it dormin and by F T Addicot and co-workers in the USA who called Abscisin II.

Effects of ABA.

- In seeds, growth of the embryo is blocked, whereas in buds and other organs, growth of the shoot apical meristem is inhibited.
- Dormancy is a complex trait in both buds and seeds. It is an important mechanism for survival during the life cycle of plants. ABA is the only hormone that induces and sustains dormancy in seeds and buds.

ii) Bud dormancy

- levels of ABA increase in leaves and buds when dormancy occurs in the short days of late summer in temperate regions. Direct application of ABA to non-dormant buds causes dormancy.
- These observations indicate that ABA is a dormancy hormone. It is synthesized in the leaves and translocated to buds where it induces dormancy.

iii) Seed dormancy

Exogenous (not synthesized in the plant) ABA inhibits seed germination in some species.

ABA levels decreases in seeds as dormancy is broken.

Functions of Absciscic acid

Stimulates closing of stomata in the epidermis.

Helps in the maturation and development of seeds.

Inhibits plant metabolism and seed germination.

It is involved in regulating abscission and dormancy.

It is widely used as a spraying agent on trees to regulate dropping of fruits.

Induces seed-dormancy and aids in withstanding desiccation and various undesired growth factors.

10.4 Summary

In this lesson we looked at the five classes of plant hormones. We saw that they have varying effects on the growth and development of plants. Ethylene is a gas. Absciscic acid is an inhibitor.

10.5 Questions.

1. What do you understand by the term plant hormone?
2. Distinguish between basipetal and acropetal transport.
3. What do you understand by the term apical dominance?
How can you demonstrate that it occurs as a result of hormone effects?
4. What is meant by epinasty?